



A review on state of health estimations and remaining useful life prognostics of lithium-ion batteries

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ABSTRACT

Lithium-ion batteries have been generally used in industrial applications. In order to ensure the safety of the power system and reduce the operation cost, it is particularly important to accurately and timely estimate the state of health (SOH) and predict the remaining useful life (RUL) of lithium-ion batteries. With the development of intelligent tools such as artificial intelligence, big data analysis and the Internet of Things, the methods of battery health assessment have been gradually diversified. Here, we have compiled four publicly available battery datasets. The SOH estimations and RUL prognostics of lithium-ion batteries are reviewed by analyzing the research status. To this end, after studying different scientific and technical literatures, the respective methods are divided into specific groups, and the advantages and limitations of the battery management system application are discussed. At the end, the future development trend and research challenges are analyzed. All key insights in this review will hopefully drive the development of battery health estimation and life prediction techniques.

1. Introduction

With a view to coping with the dual pressure in the field of environment and energy, all countries in the world regard the exploitation and utilization of new energy as a significant strategy for sustainable development. Coal, oil, natural gas and other conventional energy sources are the main energy sources in modern society. With the increasing demand for energy, it will lead to the increasing shortage of conventional energy and thus restrict the sustainable development of human society. In order to ease the energy tension and promote the sustainable development of society, it is necessary to actively develop energy conservation and emission reduction technologies and continuously explore diversified energy sources. In recent years, with the gradual improvement of battery technology, the equipment with battery as the main energy supply or energy storage component is widely applied. It is precisely because of the advantages of high output voltage, long cycle life, large energy density, low self-discharge rate, and wide operating temperature range, lithium-ion batteries (LIBs) are often widely used in portable terminals, electric vehicles (EVs), aerospace and other fields [1–4]. Tesla EV, for instance, is using the 18650 LIBs provided by Panasonic as its power source, as shown in Fig. 1.

However, there are many safety problems in the use of LIBs. Battery failures can give rise to performance degradation or failure of electrical equipment/systems, or even mission failure. For example, the improper management of LIBs in electric vehicles can cause fires and explosions [5]. In the aerospace field, one of the main causes of mission failure is power system failure [6,7]. As one of the effective methods to solve the challenge of difficult recovery and disposal for retired batteries, cascaded utilization of LIBs has attracted wide attention of researchers. For the cascaded utilization of LIBs, the safety of battery is a primary consideration. Therefore, an intelligent and high-efficiency battery management system (BMS) must be equipped for LIBs to effectively manage their safety and reliability. BMS in practical application requires the coordination of software and hardware technology. Some manufacturers have done a lot of research on the key technologies of BMS, and some of the more representative results are shown in Table 1. Detailed application of BMS can be referenced in [8].

Accurate and timely SOH estimation and RUL prediction are critical to BMS. Fig. 2 shows the relationship between SOH and RUL of the battery. Through estimating SOH and predicting RUL, the performance of each battery can be understood, and the battery life information can be obtained in advance, so as to ensure the safe and credible operation of power system or battery pack. The accuracy of the RUL prediction

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Nomenclatures

ANN	Artificial neural network
RMA	Autoregressive moving average
ARIMA	Autoregressive integrated moving average
AR	Autoregressive model
BMS	Battery management system
CV	Cyclic voltammetry
CC-CV	Constant current and constant voltage
CPR	Capacity regeneration phenomenon
DCSP	Direct current short pulse
DVA	Differential voltage analysis
DEKF	Dual extended Kalman filter
EV	Electric vehicles
EIS	Electrochemical impedance spectroscopy
EM	Electrochemical model
EMD	Empirical mode decomposition
ECM	Equivalent circuit model
EKF	Extended Kalman filter
FL	Fuzzy logic
FIM	Fractional impedance model
FBM	Fractional Brownian motion
FOA	Fruit-fly optimization algorithm

GPR	Gaussian process regression
HPPT	Hybrid power pulse tests
ICA	Incremental capacity analysis
KF	Kalman filter
LIB	Lithium-ion battery
LSTM	Long short-term memory
OCV	Open circuit voltage
PDEs	Partial differential equations
P2D	Pseudo-two-dimensional
PDF	Probability density function
PF	Particle filter
RUL	Remaining useful life
RVM	Relevance vector machine
RNN	Recurrent neural network
SOH	State of health
SPM	Single particle model
SOC	State of charge
SEI	Solid electrolyte interphase
SVM	Support vector machine
SVR	Support vector regression
UKF	Unscented Kalman filter
UPF	Unscented particle filter
WP	Wiener process

methods will directly affect the overall performance of the BMS. As a consequence, there are more and more researches on SOH and RUL of LIBs. Publications about SOH estimation and RUL prediction of LIBs according to the search results from the Web of Science have been counted. As shown in Fig. 3, the statistics of the results published in the past years show that researches on the SOH estimation and RUL prediction of batteries are receiving more and more attention.

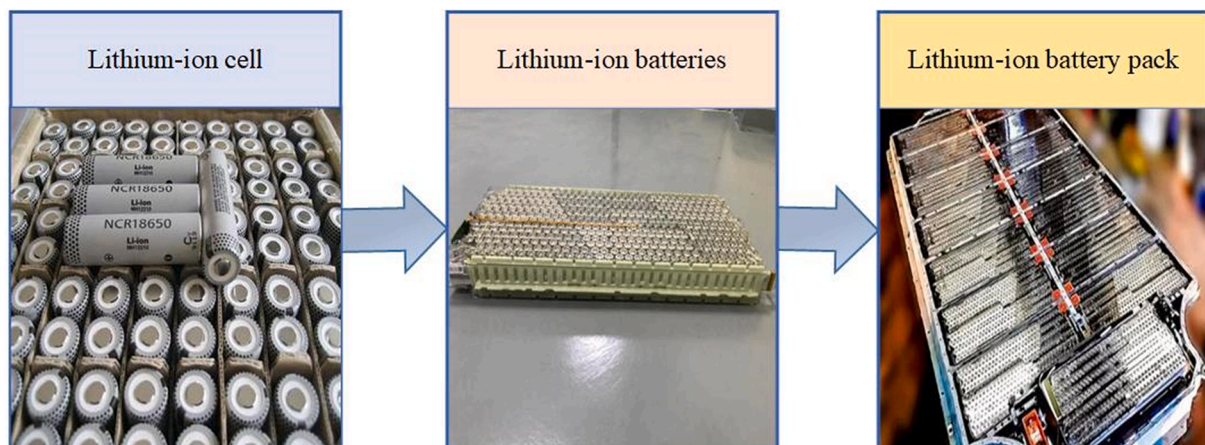
A review of studies related to the health management and life prediction of LIBs in previous years includes: Shen et al. [8] reviewed the research of multiphysics model simulation and multifunction integrated BMS technology. However, this paper focuses more on summarizing the battery multiphysics model simulation. Tian et al. [9] classified and discussed the SOH estimation method of cell management system in photovoltaic system. It has to be said that the discussion of data-driven approaches is not in-depth enough, and the discussion of the challenges and future development is one-sided. Berecibar et al. [10] proposed a classification of battery SOH estimation method, but it wasn't comprehensive. Cuma et al. [11] reviewed the estimation strategies and methods used in electric vehicles. However, the article lacks the evaluation and comparison of the investigated work, which is extremely

Table 1

Overview of the representative BMS [8].

Manufacturers	Functions
Tesla Motors	1. Acquisition and estimation of voltage, current, SOC and insulation signals; 2. Thermal management and safety protection system; 3. Low cost passive equalization.
Axeon UK Ltd	1. Recording data; 2. Health status diagnosis and residual life prediction; 3. Alert function and transmitting information to the host.
Bosch Rexroth Group	1. Startup or initialization judgment; 2. Acquisition of data and remoting transmission signals.
Toyota Motor Corporation	1. Provide demand information on battery maintenance and replacement; 2. Restrain battery imbalance.

important. Moreover, a large number of new studies have been published so far, and the real-time performance of the above two papers need to be updated. Xiong et al. [12] provided a detailed introduction of SOH estimation algorithm for batteries and emphasized its limitations and advantages, but the authors didn't give an enough discussion of the

**Fig. 1.** The application of LIBs in Tesla.

challenges and prospects. Lipu et al. [13] briefly summarized SOH and RUL estimation of LIBs in EVs. However, the research work of classification methods has not been fully summarized and discussed in depth. Li et al. [14] mainly reviewed health estimation and prognostics of LIBs based on data-driven methods. The biggest limitation of this article is that it provides only a summary of the data-driven approach, which is clearly not sufficient for beginners. Hu et al. [15] provided a concise and easy-to-understand overview of existing methods, key issues, technical challenges and future trends in the field of battery state estimation from six perspectives. This article focuses on so many points that the author doesn't investigate each point deeply enough. Tian et al. [16] reviewed the SOH estimation method in detail and put forward corresponding recommendations. Although the authors give the definition and relationship of SOH and RUL, only the relevant work of SOH estimation is investigated in detail, and there is no investigation of RUL prediction. Hu et al. [17] made a detailed classification of the battery RUL prediction, but the authors didn't discuss relevant studies on SOH estimation. Ungurean et al. [18] conducted a structured review of the models, algorithms, and commercial devices used to estimate battery SOH/RUL. However, with the development of battery research work, the work summarized in this article is not new enough. Sarmah et al. [19] conducted a comprehensive literature study on the advances, challenges, concerns and future aspects of battery SOH estimation, but the investigation of relevant work was clearly insufficient and no comparison was made with existing work. These review studies are summarized in Table 2.

As highlighted in Table 1, recent reviews of SOH estimations and RUL projections have several major limitations. In general, some of the current literatures are incomplete or one-sided, and some lack evaluation and discussion of methods. Therefore, it is necessary to make up-to-date and more comprehensive review of SOH estimation and RUL prediction of LIBs. In view of the limitations of previous literatures, this paper aims to achieve the following objectives:

- (1) The development of SOH estimation is divided into two parts: direct measurement and indirect analysis. Among them, it is mainly introduced from the two aspects of direct measurement and model-based methods.
- (2) Based on the classification of SOH, the latest research progress of RUL in recent years is discussed in detail from the perspectives of data-driven and hybrid methods.

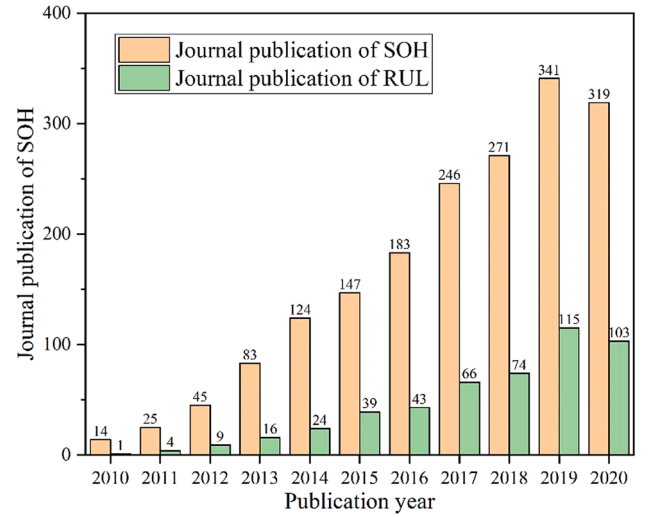


Fig. 3. Publications of SOH estimation and RUL prediction for LIBs in the recent decade from Web of Science.

- (3) The strategies and results of the summarized cases for SOH estimation and RUL prediction are illustrated in a comparative manner.
- (4) A roadmap for intelligent battery health management is described, which includes potential research trends and provides valuable guidance for researchers in future work.

In addition, this paper investigated more on the SOH estimation and RUL prediction methods, rather than empirical SOH estimation and RUL prediction from the engineering viewpoints.

The remainder of this paper is organized as follows. Section 2 describes the currently available battery datasets. Section 3 summarizes the various SOH estimation methods used by the researchers. In Section 4, various methods of RUL prediction are introduced. In Section 5, the methods reviewed above are compared and analyzed. Section 6 outlines some challenges and recommendations for future research. Section 7 concludes the paper.

2. Publicly available battery datasets

LIB is a complex nonlinear system and its performance degradation is

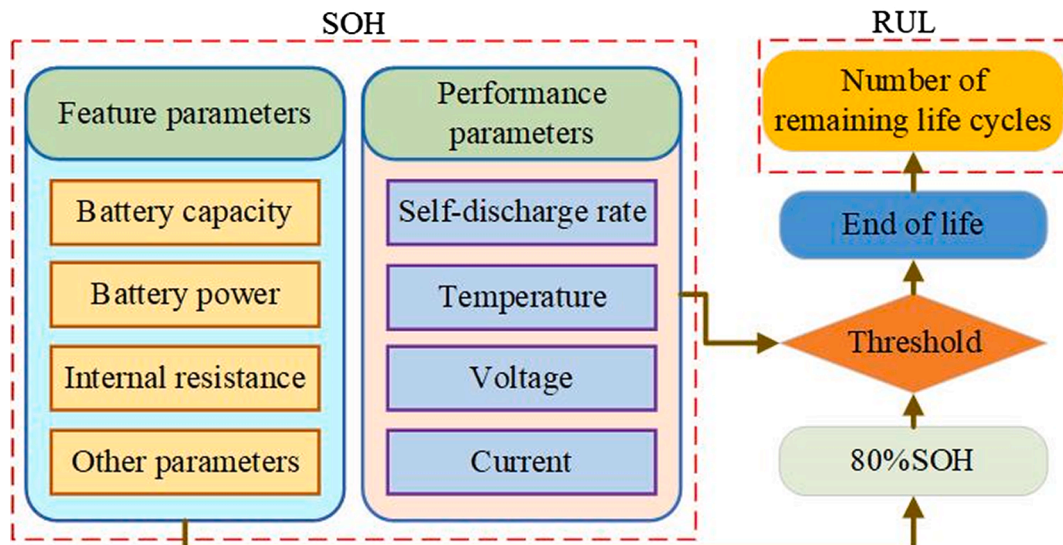


Fig. 2. The relationship between SOH and RUL.

Table 2

A summary of the published literature on SOH/RUL prediction for LIBs.

Refs.	Content	Limitations
[8]	Review on BMS from the modeling works to its multiapplication and integration.	The emphasis is on multi-physical model simulation. There are few investigations on SOH estimation.
[9]	Review of SOH estimation techniques in photovoltaic system.	Data-driven methods aren't discussed in depth.
[10]	Review of SOH estimation approaches and diagnosis tools of ageing mechanism.	Although the discussion of SOH estimation methods is detailed, the classification may not be comprehensive enough.
[11]	Review of the estimation strategies and methods used in EVs.	It is only a summary of estimation strategy, without corresponding evaluation and recommendations.
[12]	Review of SOH estimation methods.	Current problems and recommendations may not be comprehensively discussed.
[13]	Review of SOH estimation and RUL prediction approaches for LIBs.	The classification of methods is not clear and the discussion of each class of methods is not comprehensive.
[14]	SOH estimation and RUL prediction methods are reviewed from the data-driven perspective.	Only the data-driven approaches are discussed.
[15]	General review on state estimation for battery management from six perspectives.	There are so many perspectives discussed that the estimation methods of each perspective are not comprehensive enough.
[16]	SOH estimation by roughly focusing on model-based, data-driven and hybrid methods.	The definition of RUL and its relationship with SOH are mentioned, but the RUL prediction methods are not summarized.
[17]	General review on RUL prediction methods.	The SOH estimation methods are not summarized.
[18]	Review of SOH estimation in the context of embedded applications.	The works summarized in this paper are quite old and the RUL prediction work does not be adequately summarized.
[19]	General review on SOH estimation of energy storage systems.	The investigation is inadequate and no comparison of relevant works are made.

related to many factors. Both SOH estimation and RUL prediction of batteries need to be based on battery aging data. With the rapid development of battery technology, the battery RUL has been continuously improved. Sufficient degradation data can be obtained by periodic testing of the battery. For instance, the accelerated battery aging test can obtain degradation data under different conditions (i.e. charging and discharging rate, environment, temperature) in a short time. However, battery aging testing is a time-consuming and costly process. Most researchers do not have the conditions to test for battery aging, so public datasets play an important role. It helps to develop the SOH estimation and RUL prediction approaches. Many researchers have used these open datasets to study various algorithms of SOH estimation and RUL prediction. A summary of the four currently available battery datasets are provided in Table 3.

Table 3

An overview of the publicly available battery datasets.

Dataset	Refs.	Description
Experimental dataset on accelerated aging of lithium batteries	[20]	LIBs have been operated under 3 different operating conditions, including charging, discharging, and EIS.
Full life cycle test data of fuel cells	[23]	Two types of data have been provided: polarization and EIS parameters, and aging parameter data.
CS2 dataset	[25,26]	It includes 6 types of experimental data from batteries that all have undergone the same CC-CV charging process.
Randomized Battery Usage Data Set	[27]	The dataset includes data under 7 different experimental conditions.

(1) Experimental dataset on accelerated aging of lithium batteries.

NASA Ames Prognostics Center of Excellence [20] designed a series of lithium battery accelerated aging experiment under the different environmental parameters and usage. Six datasets were provided for RUL prediction. In the experiment, the lithium battery was operated under three different operating conditions: charging, discharging and electrochemical impedance spectroscopy (EIS), as shown in Table 4. In different experiments, the operating environment temperature and discharge speed of lithium battery are different. After each charge and discharge cycle, based on EIS test, impedance was measured and internal parameters characterizing battery aging were obtained. The repeated charge-discharge cycle accelerated the aging rate of the lithium battery, and the experiment ended at the end of its life cycle (i.e., the capacity was less than 70% of its rated capacity, for example, from 2 Ah to 1.4 Ah). This dataset is often used for RUL prediction. For example, Liu et al. [21] and Penna et al. [22] used this dataset to verify the proposed RUL prediction method.

(2) Full life cycle test data of fuel cells.

As a promising new energy source, life prediction and health management of fuel cells has also become a hot research topic. This dataset, provided by the Fuel Cell Laboratory (FCLAB) research consortium (FR CNRS 3539, France) [23], includes data related to the temporal and frequency domains. It plays a key role in developing SOH estimation and RUL prediction algorithms for PEMFC fuel cells. A recent study [24] used this dataset to estimate SOH of fuel cells. The experimental platform built by FCLAB to measure the dataset is shown in Fig. 4.

The operating conditions of the fuel cell (FC) can be grasped as precisely as possible by measuring and controlling several physical parameters involved in the stack, as shown in Table 5. While controlling and collecting health monitoring data, normal and accelerated aging experiments were carried out on fuel cell system under constant or varying operating conditions. Among them, the experimental data of FC1 group under stable working conditions was used as training data. The experimental data of FC2 group under the dynamic current condition was taken as test data, and this group of data was only cut off to the data of $t = 550$ h.

(3) CS2 dataset.

CS2 is a battery dataset offered by the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland [25,26]. Like the first dataset mentioned above, this dataset has also been used to validate the proposed health state estimation algorithm and life prediction algorithm in many studies. The CS2 battery dataset includes 6 types of experimental data, all of which have been operated under the same constant current-constant voltage (CC-CV) charging condition, but the discharge current of them has been set differently. LIBs were first charged with a constant current of 0.55A. When the charging voltage reached 4.2V, it started to charge LIBs with a constant voltage, and

Table 4

Experimental data under three different operating conditions.

Operation type	Data of experimental process
Charge	1. Terminal voltage; 2. Output current; 3. Temperature; 4. Current measured at charger; 5. Voltage measured at charger; 6. Time.
Discharge	1. Terminal voltage; 2. Output current; 3. Temperature; 4. Current measured at load; 5. Voltage measured at load; 6. Time; 7. Capacity at the end of discharge (2.7 V).
EIS	1. Current in sensor branch; 2. Current in battery branch; 3. Ratio of currents; 4. Battery impedance; 5. Calibrated and smoothed battery impedance; 6. Estimated electrolyte resistance; 7. Estimated charge transfer resistance.

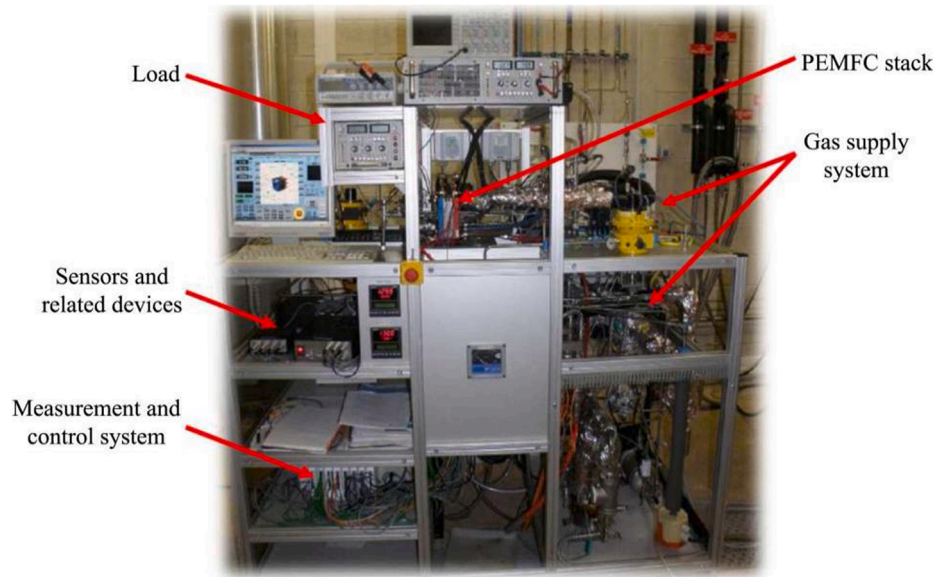


Fig. 4. Experiment platform in FCLAB [23].

Table 5

Range of physical parameters controlled [23].

Parameter	Control range
Cooling temperature	20 °C to 80 °C
Cooling flow	0 to 10 L/min
Gas temperature	20 °C to 80 °C
Gas humidification	0 to 100% RH
Air flow	0 to 100 L/min
H ₂ flow	0 to 30 L/min
Gas pressure	0 to 2 bars
Fuel Cell current	0 to 300 A

finished charging when the charging current reached 0.05A. Under different discharge rate, voltage was reduced to the specified discharge cut-off voltage. When the charging capacity of the battery reached about 80% of the rated capacity, its life was considered to end and the cycle life experiment was over. The cyclic life experiment dataset information is shown in Table 6.

Table 6

Information of CS2 battery data set.

Group	Included battery	Charge current	Discharge current	Discharge voltage range
1	CS2-8, CS2-21, CS2-33, CS2-34	0.55A	0.55A	4.2 V-2.7 V
2	CS2-35, CS2-36, CS2-37, CS2-38	1.1A	0.55A	4.2 V-2.7 V
3	CS2-3, CS2-9	Each discharge cycle alternated among 0.11A, 0.22A, 0.55A, 1.1A, 1.65A and 2.2A	0.55A	4.2 V-2.7 V
4	CS2-7	0.55A	0.55A	Discharge starts at 4.2 V, and the discharge cut-off voltage is random.
5	CS2-5, CS2-6	0.55A	0.55A	3.77 V-2.7 V
6	CS2-24, CS2-25	0.55A	0.55A	4.2V-3.77 V

(4) Randomized battery usage dataset.

Lithium batteries were continuously operated with different charging and discharging currents, where such kind of continuous charging and discharging operation is called random walk (RW) operation. For instance, a set of 18650 LIBs (RW9, RW10, RW11, and RW12, respectively) were continuously operated under a sequence of charging and discharging currents between -4.5A and 4.5A. The data set was provided by NASA Ames Research Center. It contains 7 groups of data under different experimental conditions, as shown in Table 7. This data set has been widely used in SOH estimation research. For example, Fan et al. [28] analyzed the degradation characteristics of LIBs through this data set and selected the observable variables for SOH estimation.

3. SOH estimation methods

The degradation process of LIBs will be accompanied by a variety of performance characteristic parameters, such as the maximum available capacity of the attenuation, the increase of internal impedance, power decline, and so on. Based on the changes of these characteristic parameters, SOH is defined as the ratio between the state value of the performance characteristic parameter for LIBs and the initial state value of the parameter under the current use state. SOH is a quantitative value used to characterize the overall aging degree of batteries. Ideally, the new battery SOH can be considered to be 100%. With the increase of

Table 7

Experimental information for 7 groups of data.

Group	Included battery	Experimental conditions
1	RW1, RW2, RW7, RW8	Experimental dataset of RW discharge at room temperature with variable charging time.
2	RW3, RW4, RW5, RW6	Room temperature RW discharge dataset.
3	RW9, RW10, RW11, RW12	Room temperature RW charge-discharge dataset.
4	RW17, RW18, RW19, RW20	Right skewed RW discharge dataset.
5	RW13, RW14, RW15, RW16	Left skewed RW discharge dataset.
6	RW21, RW22, RW23, RW24	Left skewed RW discharge dataset at 40C.
7	RW25, RW26, RW27, RW28	Right skewed RW discharge dataset at 40C.

cycle times, the accelerated aging of battery leads to the attenuation trend of SOH. When SOH drops to the specified failure threshold, the battery is considered to be ineffective. According to the different performance characteristic parameters selected, two commonly used definitions of SOH are given: one is defined from the perspective of capacity, and the other is defined from the perspective of internal resistance. The details are shown in the following two formulas:

$$SOH = \frac{Q_C}{Q_{new}} \times 100\% \quad (1)$$

$$SOH = \frac{R_{EOL} - R}{R_{EOL} - R_{new}} \times 100\% \quad (2)$$

where Q_C and Q_{new} respectively represent the maximum usable capacity of the battery and the initial rated capacity of the new battery under the current cycle C , R is the actual internal resistance under the current cycle, R_{new} represents the initial internal resistance of the new battery, and R_{EOL} represents the internal resistance at the end of the battery life.

As the number of battery cycles increases, the maximum available capacity decreases and the internal resistance increases. It is generally believed that battery failure occurs when the internal impedance increases to twice the initial impedance (i.e. $R = 2 \times R_{new}$). Similarly, when capacity is used as the characteristic parameter of the battery performance, it is stipulated that the maximum usable capacity decays to 80% of the rated capacity (i.e. $Q = 80\% \times Q_{new}$), then the battery need to be replaced [29]. The above are the two most commonly used methods for defining SOH. In addition, performance parameters such as power and number of charge and discharge cycles can also be used to define SOH. Although SOH is a function defined by these parameters, it is usually expressed in terms of capacity for simplicity [30]. The accurate estimation of SOH is of great significance for predicting the reliability and power transmission capacity of the battery and the normal operation of the system [31,32].

Researchers use many different methods or models to estimate SOH. In this paper, the methods of SOH estimation are summarized into two categories: direct measurement and indirect analysis, as shown in Fig. 5.

3.1. Direct measurements

The direct measurement method is to obtain the value of the characteristic parameters in the current state by measuring the standard aging characteristic parameters (such as capacity Q , internal resistance R , cycle times T , etc.) of the LIBs, and then calculate the quantitative value of SOH according to the definition of SOH to evaluate the degradation degree of the battery. In other words, based on the knowledge about the impact of the major parameters that affect the battery health status, the direct measurement method directly measures the battery health state indicators through internal resistance measurements, impedance measurements and other tests to estimate the battery SOH. This method is applicable to different materials and types of LIBs. However, the measurement is mostly carried out offline, which is not applicable to online real-time estimation of SOH. The direct

measurement method is classified below, and its application and evaluation results are summarized in Table 8 in details.

3.1.1. Internal resistance measurement

The length of battery life is directly affected by changes in its capacity and internal resistance. Since the internal resistance of the battery system plays a decisive role in the available power, energy efficiency and heat generation, the accurate measurement of the internal resistance is the key to SOH estimation [38]. This section describes the definition of internal resistance, and lists several methods of measuring internal resistance and their applications. Meanwhile, the research on the factors affecting the measurement of internal resistance is discussed.

The battery internal resistance has been defined according to the change of voltage curve when discharging current is applied [33], and the ohm's law could be used to approximate the battery internal resistance as follows:

$$R = \frac{\Delta U}{\Delta I} \quad (3)$$

where ΔI and ΔU stand for the step change of current and voltage, respectively. Measuring the increase of cell resistance is a direct tool for estimating SOH [34]. For instance, joule effect and hybrid power pulse tests (HPPT) can be used to measure the internal resistance. Among them, the HPPT method can determine the dynamic power capability of the device within the range of available charge and voltage [10]. Some authors have carried out related studies on measuring the internal resistance of batteries [35], so as to evaluate the subsequent health status through these measurements.

The current pulse methodology is usually applied to estimate the internal resistance. It is to make the lithium battery perform pulsed current discharge on the load through controlling the on-off of the switch tube, and to estimate the internal resistance of the battery by collecting the voltage and current before and after the discharge. The authors in Refs. [36] estimated the internal resistance by using three different timing pulses (10 ms, 2 s and 30 s). The pulse discharge method is a commonly used method to detect the internal resistance of the battery, but the internal resistance under load cannot be accurately obtained because the pulse time is in seconds. Based on this, a direct current short pulse method (DCSP) was proposed to measure the internal resistance, and the capacity was calibrated by CC-CV charge and discharge test [37]. And then, the relationship between these two parameters was established based on linear fitting. It is found that this method can accurately measure the internal resistance when the battery loads current changes. A newly developed technique using pulse-multisines (PM) to measure the internal resistance was proposed in literature [38]. The PM method used can not only estimate the non-parametric internal resistance and its standard deviation, but also estimate the ECM parameters by using weighted nonlinear optimization. Therefore, this method has strong robustness to the battery dynamic uncertainty.

Moreover, the joule effect is a law that quantitatively describes that

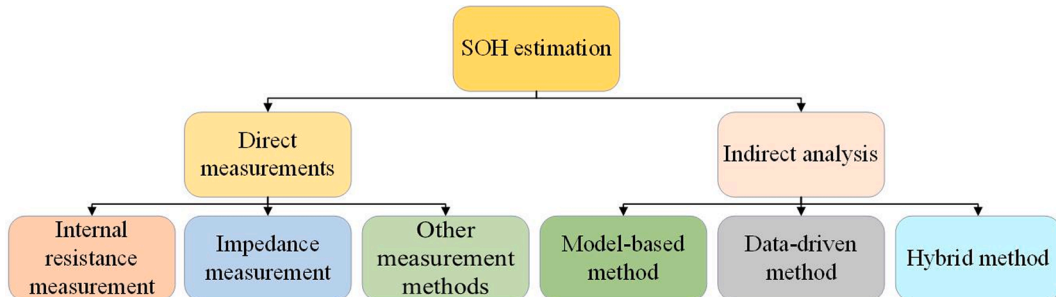


Fig. 5. Classification of SOH estimation methods.

Table 8

The applications and evaluation results of direct measurement for SOH/SOC estimation.

Methods	Refs.	Computational complexity	Evaluation index	Estimation precision
Internal resistance measurement	[39]	Adaptive algorithm and model	Maximum estimated error	0.1 V (OCV) / 0.5% (DoD or SOC)
	[37]	Function expressions, Model	RMSE	10% (fixed capacity), 1.5% (fitting capacity)
	[42]	the Thevenin model, RLS algorithm and linear equations	MAE, RMSE, Variance	Within $\pm 4\%$
Impedance measurement	[48]	Function expressions and model	Absolute error	Less than 15%
	[49]	Exponential expressions, equation set	RMSE	< 2.48%
	[51]	Grünwald-Letnikov definition, the least-squares genetic algorithm, model	Error rate	an error margin of [1.5%, 1.5%]
	[52]	The discrete state-space equations, model	MAE, RMSE	within the range of 3%
Other measurement methods	[55]	The nonlinear least square method, model	RMSE	< 0.12% (Correct initialization); < 20.1% (Incorrect initialization)
		ampere-hour counting		
	[70]	Acoustic model, function expressions	RMSE	Combined effect: 0.69% (SOC), 0.0051% (SOH)
	[71]	Acoustic model, function expressions	N/A	N/A
		ultrasonic measurement		

conduction current converts electrical energy into thermal energy. The thermal energy generated by the current passing through the conductor is proportional to the square of the current, the resistance of the conductor, and the energization time. Therefore, the joule effect technique can be used to measure the internal resistance. The battery temperature varies with the loss of this effect, and a calorimeter is often used [10]. In literature [39], an infrared radiation (IR) measurement and adaptive technology for identifying the internal resistance and open circuit voltage (OCV) was proposed. The simulations and experiments showed that the proposed method had good robustness and precision.

The internal resistance varies with the battery size, chemical composition, temperature and other parameters. It is more accurate to measure the internal resistance by taking into account a variety of factors. In literature [40], a method for estimating the internal resistance of LIBs was presented, which considered SOC, current rate and temperature. The constant current was tested at different temperature. It is concluded that SOC does not seem to be related to the increase of internal resistance. Similarly, another study also analyzed the growth of internal resistance at different SOC and different temperature [41]. It is found that the influence of SOC on the growth of internal resistance is not obvious, but the temperature has a great influence on the growth of internal resistance. Some studies estimate SOH by establishing the internal relationship between internal resistance and capacity. Among them, the internal resistance at the beginning of the battery life (R_{NEW}) and the internal resistance at the end of the life (R_{EOL}) are two more important parameters. However, since the aging process of the battery is irreversible, these two parameters are very difficult to obtain. To solve this problem, Chen et al. [42] established a connection equation for calculating R_{NEW} and R_{EOL} based on the established linear relationship between internal resistance and capacity attenuation. The R_{NEW} and R_{EOL} were estimated through two random discharge cycles in different degradation stages. In addition, establishing the relationship between the resistance and impedance of the battery also provides a method for SOH estimation [43–46].

Although the internal resistance measurement method can be applied online, because the ohmic internal resistance is small and the change with charge and discharge cycles is weak, an accurate measurement circuit is required.

3.1.2. Impedance measurement

To determine SOH, it can also be achieved by estimating the actual value of impedance. For this reason, electrochemical impedance spectroscopy (EIS) is commonly applied. EIS is a method that allows high-precision impedance measurement of batteries to be obtained in a large frequency range with low current and uses a small sinusoidal alternating current or voltage as an interference signal to test the impedance change process of the system [18,50]. EIS can measure battery impedance in a wide frequency range. Therefore, not only ohm

resistance can be obtained, but also other parameters such as charge transfer resistance and solid electrolyte interphase (SEI) resistance can be obtained [47]. This section details the application of EIS to SOH estimation and analyzes the advantages and limitations of this approach.

Based on EIS, electrochemical processes such as film formation and charge transfer at the electrolyte interface can be studied. However, different temperatures and SOC can affect electronic/ionic conduction and lithium-ion insertion/extraction reactions [48], so it is necessary to consider the effects of these factors when using EIS to estimate aging state. For example, the relationship between EIS of the LIBs and temperature and SOC was explored in literature [48]. The EIS at different temperatures, SOC and aging cycles are shown in Fig. 6. On this basis, a method for estimating battery SOH considering temperature and SOC was proposed, which is to attain the charge transfer resistance through fitting the impedance spectrum with the equivalent impedance model so as to estimate SOH. In literature [49], a capacity degradation model was established by the electrolyte impedance and charge transfer impedance to obtain the battery health condition parameters, which consisted of impedance aging parameters and capacity degradation parameters.

The impedance-based estimation methods have two disadvantages: on the one hand, the complete EIS measurement is usually time-consuming; on the other hand, battery information needs to be extracted through complex calculations. In view of the above shortcomings, a multi-point impedance technology for monitoring and screening of LIBs was proposed in literature [53]. This technology could not only distinguish ohmic resistance, contact resistance, and SEI resistance, but also it was easy to calculate. Typically, a traditional EIS is obtained under steady state and requires a long pause time. For this reason, a dynamic EIS (DEIS) was developed [54]. After measuring the impedance within the charge and discharge cycle, comparing the test results of DEIS and EIS, it is found that the most obvious characteristic of DEIS is the reduction of charge transfer resistance within the SOC range of 0.4 to 1.

In addition, a fractional impedance model (FIM) combining fractional order theory and EIS theory has been developed for SOH estimation of batteries. This model not only makes up for the lack of physical and chemical significance of ECM model, but also ensures the balance between model complexity and computational efficiency. A recent example studied the health state estimation of LIBs based on simplified FIM and battery spacing capacity [51]. Firstly, a simplified FIM based on the definition of Grünwald-Letnikov was developed. Then, the least squares genetic algorithm was applied to identify the model parameters. Based on the identified model parameters and the interval capacity, the back-propagation algorithm was introduced to estimate SOH. The experimental results verified the availability of the method. Similarly, a fractional order impedance model based on EIS data was established and derived the discrete state space equation of the impedance model through the definition of Grünwald-Letnikov [52]. The polarization effect of the cell can be described simply and effectively by the

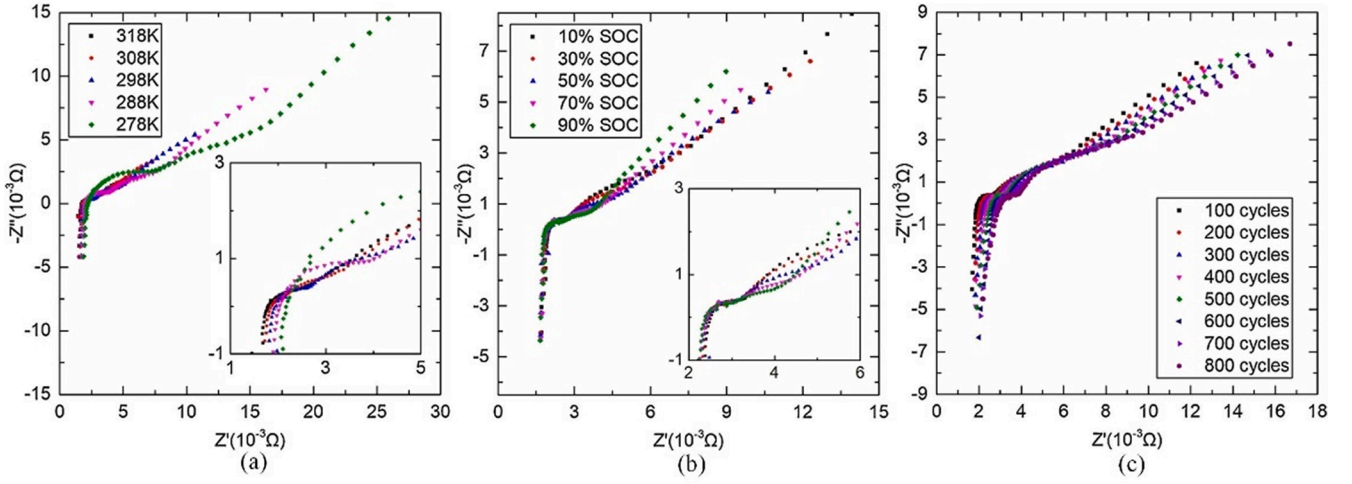


Fig. 6. The relationship between EIS of the LIBs and temperature and SOC. (a) EIS obtained under different temperature, 50% SOC after 100 aging cycles; (b) EIS obtained under different SOC, 298 K after 100 aging cycles; (c) EIS obtained under 50% SOC and 298 K after different aging cycles. [48]

fraction element.

A large number of literature studies show that the SOH information obtained by EIS analysis is effective. However, due to the cost and complexity of on-board implementation, and the EIS measurement requires a longer test time and a stable test environment, a simpler and more general method for obtaining online EIS parameters require further research.

3.1.3. Other measurement methods

The above mainly reviews the internal resistance measurement and impedance measurement. Direct measurement methods also include ampere-hour counting method, cycle number counting, destructive test, ultrasonic measurement, and so on.

The ampere-hour counting method refers to the full discharge of a fully charged LIBs, and the long-term monitoring of the battery current can obtain the maximum available capacity under the current state. For example, in literatures [55,56], the ampere-hour counting method was used to estimate SOC. However, it has to be considered that this method takes long time to test and relies on specific high-precision monitoring equipment. It is difficult to achieve online applications under actual operating environments.

The cycle counting method determines SOH by recording the number of times for the battery has discharged completely. The SOH can be calculated from number of full cycles and the number of current cycles experienced by batteries [12]. Although this method is simple and straightforward, it is only suitable for rough estimation due to the uncertainty of the whole life cycle number.

The destructive test is to observe the change of the internal microstructure for LIBs during the aging process through a specific device, so as to study the degree of battery degradation from a microscopic perspective. The most commonly used techniques include: Raman spectroscopy [57–59], X-ray diffraction [60–62], X-ray photoelectron spectroscopy [63–65], scanning electron microscope [66,67], atomic force microscopy [68,69] and so on. This method can provide accurate degradation information for SOH estimation. To study the degradation mechanism, however, the battery must be disassembled, but the damage to the battery is irreversible. Hence, this method is not applicable to actual industrial systems.

Ultrasonic detection is an emerging technology in recent years, and there are more and more related studies using ultrasonic technology to detect internal changes in the battery aging process. The reflection, refraction and wave pattern transformation will be generated at the critical interface of the cell where degradation occurs, and health diagnosis is made based on these reflected waves [12]. Compared with

existing methods, it is more compact and cost-effective for ultrasonic measurements to determine both the SOH and SOC of batteries [70]. A framework based on ultrasonic guided waves using low-profile built-in piezoelectric transducers was proposed in literature [72] to monitor the SOC and SOH of LIBs. Through time domain analysis, it is found that there is a repeatable associativity among the waveform signal parameters and SOC and SOH. Therefore, a technique for accurately monitoring the battery SOH in real time using an ultrasonic signal can be constructed through this correlation. In further work, Ladpli et al. [71] described the decrease in ultrasound acoustic time-of-flight (ToF) as the cycle increased in detail. To sum up, the application of ultrasonic testing solves the difficulty of the internal defect detection of the battery and does not need to be disassembled, so it has a good application prospect. However, this method needs further study and improvement, because the end user is difficult to grasp the battery health status from the waveform on the monitor.

3.2. Indirect analysis

During actual operation, performance parameters such as the capacity and impedance of the LIBs are not easily measured. In view of the above problems, estimation of the degradation status of LIBs based on online monitorable information has begun to be widely studied. In short, it is to find health factors that have a strong correlation with battery performance parameters and can characterize the performance degradation of battery from online monitorable information. Compared to direct measurement methods, such methods are called indirect analysis methods. This paper summarizes model-based methods, data-driven methods, and hybrid methods in indirect analysis methods. Table 9 summarizes the application and evaluation results of various methods for indirect analysis.

3.2.1. Model-based method for SOH estimation

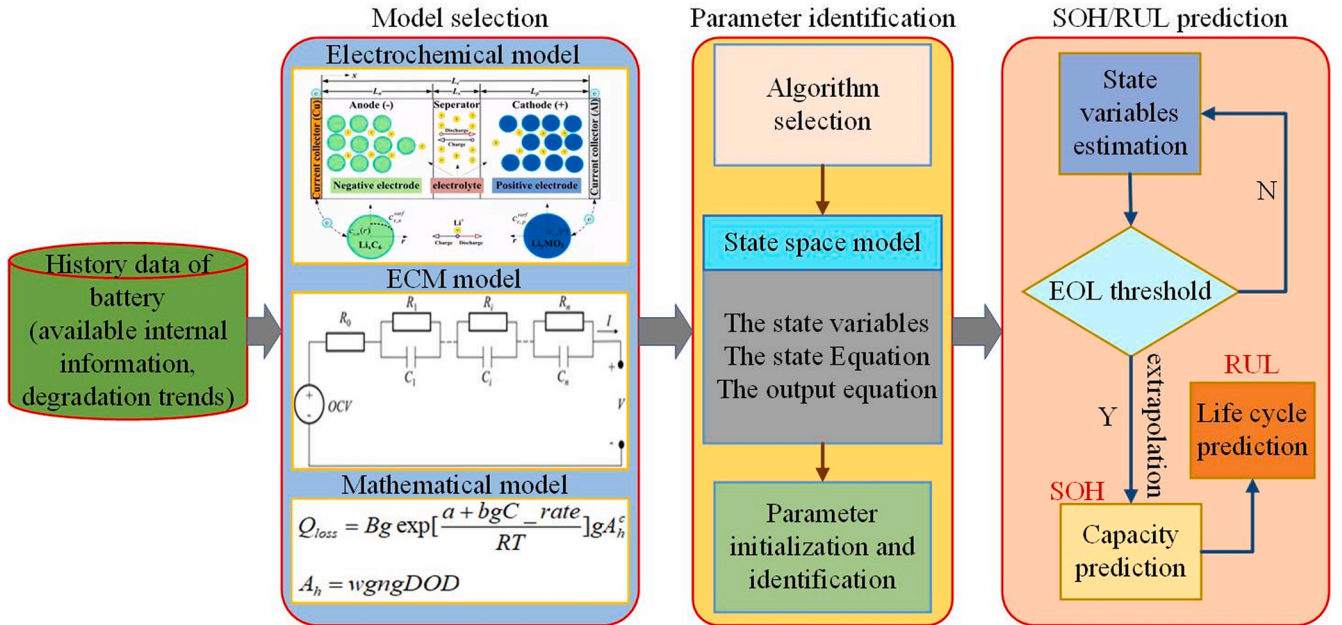
The method of estimating the SOH by modeling the measured battery performance parameters has been widely studied. As shown in Fig. 7, the appropriate model is selected first. Then the corresponding algorithm is selected to initialize and identify model parameters, and finally SOH is estimated based on the updated model. Inspired by the detailed classification of battery models in literatures [18,73], this paper summarizes model-based methods from three aspects, namely ECM-based method, electrochemical model-based method, and mathematical model-based method.

(1) ECM-based method

Table 9

The applications and evaluation results of indirect measurement for SOH/SOC estimation.

Methods	Refs.	Computational complexity	Evaluation index	Estimation precision
Model-based method	ECM-based method	[85]	Logarithmic function and Current time constant expression	Estimation error
	EM-based method	[90]	Transfer function and Moving-Window Filter	RMS error
		[93]	Matrix operation, SPM	RMSE
		[94]	Function expressions, model	N/A
	Mathematical model-based method	[101]	Exponential functions, vector dot product	Absolute error
		[102]	Polynomial functions	Estimation error
		[103]	Exponential expression and model	RMSE
		[104]	Statistics	Error
		[105]	Probability density function	Error
		[106]	Piecewise statistical model	Error
Data-driven method	ANN	[118]	Matrix operation, AST-LSTM	ARMSE
		[117]	Matrix operation, RNN, LSTM	MAE, MAEP
		[116]	Matrix operation and ANN	MRER, MAE, RMSE
		[119]	Functional operation, fuzzy logic-controlled	N/A
	Fuzzy logic-based method	[120]	The fuzzy algorithm, Nonlinear AutoRegressive model	Error
		[121]	Matrix operation, EKF	Error
		[122]	Matrix operation, KF	Mean relative error
	Bayesian filtering-based method	[114]	Matrix operation, DEKF, FFRLS	Absolute error
		[124]	Matrix operation, DEKF	Error
		[113]	Matrix operation, genetic resampling PF	N/A
		[129]	Matrix operation, UPF	AE, ME, MRE, RMSE, AWCI
	SVM	[130]	Matrix operation, SCDPF	Error
		[131]	Matrix operation, SVM	MAE
		[132]	Matrix operation, LS-SVM	RMSE
		[133]	Function expressions and SVR	RMSE
Hybrid methods	RVM	[134]	Matrix operation and RVM	RMSE
	Direct measurement + Model-based method	[135]	Double polarization model, FFRLS	Error
		[136]	FFRELS, FOM and ICA	Error
	Direct measurement + Data-driven method	[137]	Matrix operation, GM, PF	N/A
	Model-based + Data-driven	[138]	Matrix operation, BM and PF	RMSE
		[139]	Matrix operation, LS-SVM, UPF	RMSE
		[140]	Matrix operation, optimized multiple kernel RVM, UPF	AI, RMSE, MAPE
Different data-driven methods		[141]	Matrix operation, ARMA, Elman-NN	MSE

**Fig. 7.** Flowchart of the model-based method for SOH estimation.

The ECM is usually composed of the constant voltage source, resistance, inductance, capacitance and other basic elements. Fig. 8 shows a generic ECM. Through the equivalent or approximation of the dynamic

characteristics for the system, the current-voltage characteristics and the transient behavior of the battery can be captured. ECM is generally divided into two types: time-domain ECM and frequency-domain ECM

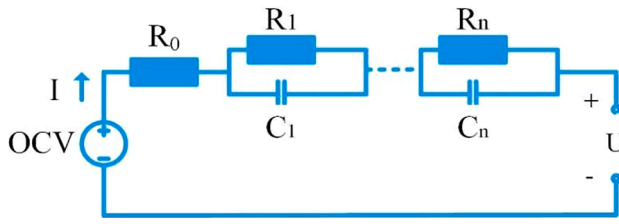


Fig. 8. A generic ECM.

[8], among which the models widely used include: Rint model [74], RC network model [75], partnership for a new generation of vehicle (PNGV) model [76], Thevenin model [77,78] and nonlinear ECM [79].

The ECM-based method first needs to identify the ECM parameters, and then SOH is estimated indirectly through the identified parameters. In literature [80], a method for parameter identification of ECM considering electrochemical performance was proposed, and the electrical parameters of ECM were calculated according to the relationship between electrochemical parameters and the internal changes of solid and electrolyte phases. Combined with the electrochemical transfer function and the equivalent circuit form, the relationship between resistance/capacitance and electrochemical parameters was established to better reveal the internal mechanism of the battery, which can be used for further researches (e.g., SOC estimation). Considering electrochemical performance as well, the authors in literature [81] proposed an electrochemical-based ECM (E-ECM). E-ECM has two more parameters than the second-order RC model, which can better simulate the voltage response of batteries. In addition, considering the balance between model complexity and accuracy and the ability to characterize the nonlinear characteristics of LIBs, a recent study proposed a new fractional variable order ECM (FVO-ECM) [82]. The RC branches in this model can make continuous order changes to obtain fractional RC branches, which can reduce the complexity of parameter identification.

Another parameter identification idea is to use the least square (LS) method and other evaluation techniques combined with measurable state signal to identify SOH-related parameters online. These methods adjust the model parameters to obtain the optimal solution based on the error between the model output and the actual output. The recursive least square (RLS) method and its improvement are the most classic online parameter identification methods. For instance, in literature [83], the RLS algorithm was worked for the online identification of ECM parameters, and the high identification accuracy of the algorithm was verified by experiments. In literature [84], the authors used a fast UD factorization-based recursive least square (FUDRLS) algorithm to identify time-varying electrical parameters. To optimize the performance, variable forgetting factor (VF) was added to the algorithm. Hence, compared with the traditional RLS-based parameter estimation method, the proposed method has lower complexity and better numerical stability.

After the ECM parameters are identified, these parameters can be used to estimate SOH. Based on the first-order ECM and the recursive form of constant-voltage (CV) charging current, the detailed expression of current time constant was derived in literature [85]. The expression is shown in Eq. (4):

$$\tau_I = \frac{C_p R_p R_O}{R_p + R_O} \quad (4)$$

where C_p stands for the polarization capacitance, R_p and R_O represent the polarization resistance and Ohmic resistance respectively. The aging state of the battery was characterized by the time constant of the CV charging current. Next, a quantitative relation between the normalized battery capacity and the current time constant was built, and the load current during the CV charging process was analyzed to estimate SOH online.

(2) Electrochemical model-based method

The electrochemical model (EM) fully considers the actual electrochemical reaction process in batteries, including the solid-phase diffusion process, liquid-phase diffusion and migration process, etc. Fig. 9 shows an electrochemical impedance model. It is a complete description of the reaction mechanism of the battery. It is composed of a series of electrode and electrolyte dynamics partial differential equations (PDEs), which has high computational accuracy but large computational amount [8]. It is worth noting that massive simplified models have been developed and widely used for SOH estimation. When the EM is used to model the health state of LIBs, the electrochemical law of battery capacity decline is mainly studied by the electrochemical reaction mechanism of the battery. The model input is the amount of electrochemistry in the battery, and the output is the decline of capacity. A simplified transfer function-type electrochemical aging model considering side reactions was developed in literature [86]. The model was built by deriving the transfer function relation between the input current and the aging representatives such as the resistance of solid electrolyte inter-phase and the thickness of the deposited layer resulted from side reactions. It is worth mentioning that this method can well bridge the gap between the mathematical modeling of battery and the research of aging mechanism. In addition, an improved EM model was proposed, which can not only save computing resources but also reduce processing time and obtain accurate results at the same time [87].

As a popular EM, the Pseudo-Two-Dimensional (P2D) model has been widely used in the research of LIBs [88]. The complex physicochemical properties of battery are described by a series of PDEs [89]. In literature [90], an EM considering the physical and chemical properties of LIBs was established to evaluate SOC. A method based on projection of the optimal basis function was proposed for solving the solid-phase diffusion PDEs in the model. To improve the convergence rate of the state filter, a new moving window filter (MWF) method was introduced.

In view of the complexity of P2D model, more and more numerical solutions with parameter identification and simplified form of P2D are developed. As a simplified model of P2D model, the single particle model (SPM) that has the advantages of simple model and small computation, has received great attention in SOH estimation and other aspects [91,92]. For example, Li et al. [93] developed an improved capacity degradation model with chemical/mechanical interpretation mechanisms that integrated advanced SPMs and could be directly applicable to the SOH estimation in a BMS. Furthermore, another research [94] proposed a modified reduced-order EM based on SPM to estimate SOH. The electrolytic phase potential difference is introduced in the improved model, and the influence of unmodeled dynamics is considered. Compared with the traditional SPM algorithm, this algorithm has significant improvement in computational efficiency and predictive performance, and can predict the battery performance under a wide range of operating current conditions.

(3) Mathematical model-based method

The influencing factors and operating characteristics of LIBs are complex. However, the application of mathematical models can transform these influencing factors and operating characteristics into mathematical relationships for analysis. In view of the fact that both incremental capacity analysis (ICA) and probabilistic methods involve

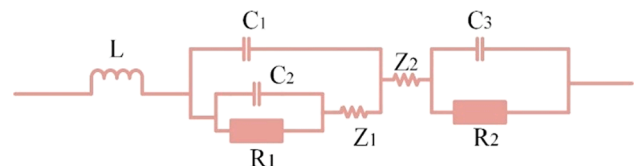


Fig. 9. The electrochemical model.

numerical analysis, the mathematical models are mainly summarized in these two categories. In addition, there are some other methods involving mathematical theory, and those interested can refer to literatures [95–97] and so on.

Initially, ICA was used to study the characteristics of intercalated batteries [98–100], but in recent years, the method has been widely used to estimate SOH. Weng et al. [101–103] conducted in-depth research on ICA-based SOH monitoring. In literature [101], the SOH was monitored by analyzing the battery cycle aging data captured by the ICA-based OCV model to identify robust characteristics related to battery aging. The OCV model was expressed as Eq. (5):

$$y(z) = K_0 - \frac{K_1}{z} - K_2 z + K_3 \ln(z) + K_4 \ln(1 - z) \quad (5)$$

where K_i ($i = 0, 1, 2, 3, 4$) and y are the model parameters and the OCV model respectively, and z is the normalized SOC. Similarly, the literature [102] also used the same method to estimate SOH. The difference was that the literature makes corresponding improvements on the basis of literature [101], and proposed an improved OCV model with the following expression:

$$y(z) = K_0 + K_1 \frac{1}{1 + e^{\alpha_1(z - \beta_1)}} + K_2 \frac{1}{1 + e^{\alpha_2(z - \beta_2)}} + K_3 \frac{1}{1 + e^{\alpha_3(z - 1)}} + K_4 \frac{1}{1 + e^{\alpha_4 z}} + K_5 z \quad (6)$$

where α_i ($i = 1, 2, 3, 4$) and β_i ($i = 1, 2$) are the nonlinear parameters, and y represents the OCV model. It had to be said that using more complex models can lead to better results. Furthermore, in literature [103], the author proposed an incremental capacity (IC) peak tracking framework based on ICA principle to monitor SOH. When a constant current is applied, the IC curve can be calculated by Eq. (7):

$$IC_K = \frac{\Delta Q_K}{\Delta OCV_K} = \frac{\Delta Q_K}{\Delta OCV_K + R_0 \Delta I_K} = \frac{\Delta Q_K}{\Delta V_K} \quad (7)$$

where ΔI_K , ΔQ_K and ΔV_K are the changes of current, charging capacity and voltage in sample time K respectively, and R_0 is the internal resistance.

There is no doubt that probabilistic methods belong to the category of mathematical models. In literature [104], a probabilistic framework for battery pack prediction was established, which bridged the gap between battery and battery pack aging prediction in a probabilistic sense. The probability density function (PDF) was applied to characterize the battery state, and the prognosis was further improved by estimating the aging model parameters of package. In literature [105], the aging rules of batteries were analyzed by a PDF method, and an online SOH estimation algorithm was proposed based on the regularity of peak height reduction and battery aging. The comparison of PDF method, cyclic voltammetry, ICA and differential voltage analysis (DVA) showed that the four methods had the same mathematical basis. The PDF method is not only simple, but also extends the application of ICA/DVA approach, so it is suitable for BMS. The capacity regeneration phenomenon (CRP) will seriously threaten SOH estimation results. In view of this factor, Wang et al. [106] used expectation-maximization algorithm to detect CRP probabilistically, and successfully improved the prediction accuracy based on the proposed piecewise statistical model.

3.2.2. Data-Driven method for SOH estimation

In general, the data-driven method is to mine the implicit battery health information from the battery historical degradation data and state monitoring data (current, voltage, temperature, etc.), and realize SOH estimation by combining machine learning or intelligent algorithm. There are many methods based on data-driven, including gaussian process regression (GPR) [107,108], wiener process (WP) [109], autoregressive model (AR) [110], particle filter (PF) [111–113], Kalman filter (KF) [114,115], etc. Compared with model method, the data-driven method has more flexibility and applicability. However, the

training time is long and the generalization ability is low. This section roughly divides data-driven methods for SOH estimation into neural network-based method, fuzzy logic-based method, Bayesian filtering-based method and support vector machine (SVM)/relevance vector machine (RVM)-based method. Section 4.1 also categorizes data-driven methods in more detail.

(1) Artificial neural network-based method

Artificial neural network (ANN) is capable of handling large amounts of data in complex nonlinear systems and is independent of the electrochemical principles inside the battery [13]. ANN contains many kinds, such as recurrent neural network (RNN), long short term memory (LSTM) neural network, back propagation neural network (BPNN), and so on. In literature [116], ANN and signal processing method were used to estimate SOH online under constant discharge state. In order to retain the characteristic values strongly related to battery aging, the Gaussian filter method was used to smooth the IC curves. Then the characteristic values were extracted from the local area of the IC curves and used as the input variable of the ANN model, while the SOH was regarded as the output variable.

A large number of studies have proved that the prediction method based on deep learning is superior to the traditional machine learning method. Among them, LSTM NN is one of the most effective approaches for the prediction of adaptive time series. For instance, considering the variable input dimension, Zhang et al. [117] estimated the battery SOH based on RNN and LSTM structure, and realized the adaptive time series prediction. Although LSTM can achieve better estimation performance, there are still some shortcomings. LSTM uses constant error carousel (CEC) that is not directly connected to prevent the rapid decay of the error signal, which will lead to a lack of learning essential information. Usually, when constructing the SOH estimation model, only capacity is considered as the input of LSTM, and the LSTM that cannot be shared by multiple batteries has limitations in practical applications. Hence, Li et al. [118] developed an active states tracking-LSTM neural network (AST-LSTM NN) that can filter more useful information. The structure of the model is shown in Fig. 10. SOH model with many-to-one structure was trained by mapping multiple parameters to the AST-LSTM.

(2) Fuzzy logic-based method

Fuzzy logic (FL) is a method of simulating human brain to implement regular inference. It can differentiate fuzzy sets and handle fuzzy relations by means of membership function (MF). The MF has a great relationship with the accurate estimation of SOH. Literature [119] studied an online SOH prediction method based on FL rule set. The maximum capacity and resistance of the battery are two key parameters in this method. These two parameters were calculated by measuring voltage, current, time and temperature under certain conditions, and then the appropriate MF was selected to predict SOH according to the established FL reasoning system. In literature [120], the authors used FL to effectively calculate the health index of the battery under different operating and environmental conditions through the fitting curve representing capacity decay and characteristics. Among them, the process of health index calculation is shown in Fig. 11.

(3) Bayesian filtering-based method

The entire family of Bayesian filters includes KF, PF and their variants, which are widely used to solve the problem of dynamic state estimation. In Bayesian inference, model parameters are updated and estimated based on observations. The appropriate filter is usually selected according to the system dynamics, shape of noise distributions and the filter itself [14].

1) **KF and its variants.** KF has advantages in dealing with problems of linear system with Gaussian noise. For example, Plett et al. [121] used

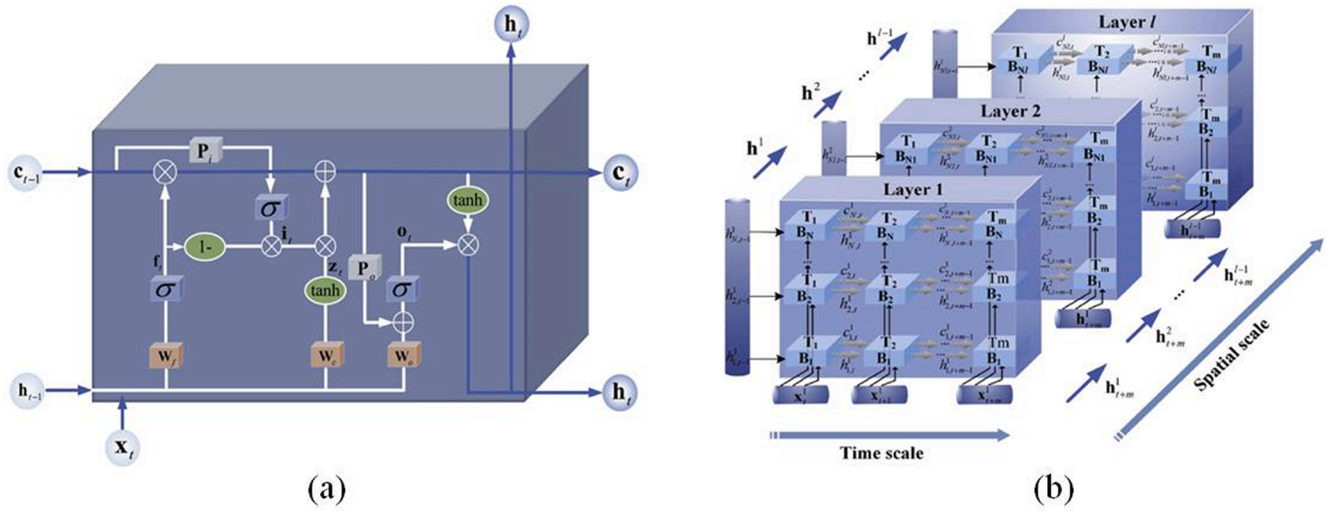


Fig. 10. The structure of AST-LSTM NN [118]: (a) The structure of cell; (b) The structure of deep NN.

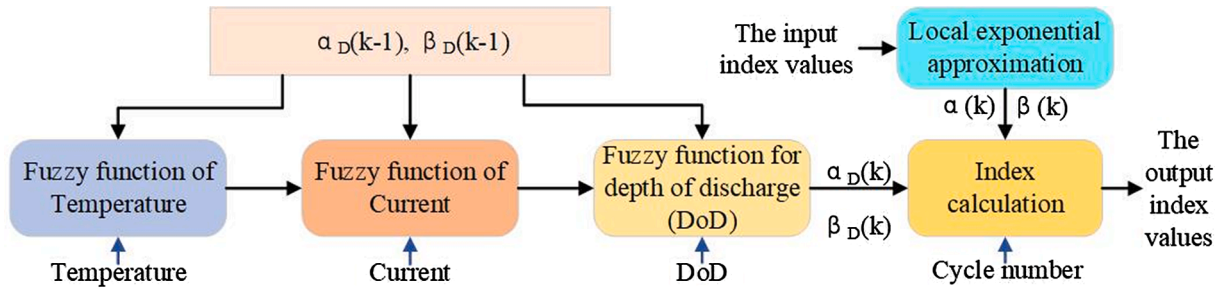


Fig. 11. Fuzzy logic controller schematic diagram ($\alpha_D(k)$ and $\beta_D(k)$ are the corrective coefficients).

KF to develop a battery model and estimate its parameters. In another example, the polarization and dynamic characteristics of the battery were described by the Thevenin model and estimated by KF [122]. However, the degradation process of batteries is usually nonlinear. Therefore, variants of KF, such as Extended Kalman Filter (EKF), dual Extended Kalman Filter (DEKF), and Unscented Kalman Filter (UKF), are developed. A recent study showed that using dual KF (DKF) algorithm to filter SOC twice can cut down the interference caused by battery model error and current measurement error system [123]. Most off-line KF algorithms cannot accurately describe the actual operating conditions of batteries, while DEKF composed of two EKFs can synchronically estimate battery parameters and states. For instance, a spatial model in the discrete status for DEKF based on the internal resistance calculated according to Thevenin model was established to estimate the battery SOH online [124]. Fang et al. [114] established ECM based on the relationship between internal resistance and SOH, and used forgetting factor recursive least-squares (FFRLS) method to identify model parameters online. Then, the SOC and SOH were estimated online jointly by DEKF algorithm. In addition, compared with EKF alone, DEKF allows the filter to compensate for static initialization of model parameters and low noise covariance due to its additional freedom. Wassiliadis et al. [115] estimated model parameters based on DEKF and found that higher model accuracy and filter robustness could also be obtained in the late degradation stage of the battery.

The model error is inevitable because the parameters of the battery model will change under different operating conditions. A recent study constructed a fuzzy controller to correct the Kalman gain coefficient, thus reducing the estimation error caused by battery degradation and model error [125]. At the same time, a Cubature KF (CKF) algorithm was proposed to estimate SOC. In its infancy, CKF has the ability to promptly

converge to the reference value by setting a small observation noise covariance, it can achieve better robustness than EKF algorithm in the state estimation of the battery.

2) PF and its variants. From the above discussion, it can be seen that KF is an effective method to deal with the state estimation problem of Gaussian noise. However, in the actual operation of the battery, the state noise and the measurement noise do not conform to the Gaussian distribution, which limits the application scope of the KF algorithm. As a sequential Monte Carlo method, PF combines Bayesian inference with important sampling [14]. PF and its variants are a better choice when dealing with problems of nonlinear systems involving non-Gaussian, which do not require large amounts of historical data and are suitable for online applications. For instance, PF can use a group of particles to approximate the arbitrary PDF of noise, and can continuously update particle positions based on Bayesian theorem to realize the posterior probability density estimation of states [126]. The OCV of a battery during charging and discharging is different, so that its OCV characteristics are not clear. For this reason, traditional methods such as KF are particularly difficult in charge state estimation, while PF can overcome this difficulty with the Monte Carlo sampling method and has the ability to represent arbitrary PDF. Schwunk et al. [112] used PF to accurately estimate the SOH and SOC of the battery, thus avoiding the establishment of a complex model for processing hysteresis. Bi et al. [113] proposed a SOH estimation method based on genetic resampling PF, which not only effectively solves the non-Gaussian problem, but also achieves good prediction performance when the accuracy of ECM is not high. Given the complex aging behavior of batteries, a single model may not be sufficient to describe the degradation process. To solve this problem, a linear model and an exponential/linear hybrid model were proposed to describe the capacity degradation behavior of the cell, and the fixed-

lag Multiple Model PF was used to perform nonlinear filtering on the cell [127]. Aiming at the problem of particle degradation and PF adaptability under complex working conditions, Liu et al. [128] proposed a hybrid filter combining cubature filter and PF, where a noise self-adjusting strategy was also introduced to make the estimator more suitable for practical application.

Unscented PF (UPF) can effectively adapt to LIBs with nonlinearity and non-Gaussian properties. Meanwhile, UPF is able to combine the latest measurement information to provide a proposal distribution closer to the true posterior distribution than PF. An online SOH estimator based on UPF algorithm was developed, which fused the mapping relationship between online HI and SOH into the state space model, and provided an uncertain representation of the estimated results thanks to its better proposal distribution [129]. In addition, PF usually takes the prior probability density as the important density function. However, there is a great difference between the prior probability density and the posterior probability density, which will threaten the correctness of the collected samples. Meanwhile, the diversity of particles decreases with the increase of iteration times. In order to solve these problems, a recent study developed an algorithm based on second-order central difference PF (SCDPF) [130]. Based on the dual exponential model, the algorithm uses the second-order central difference Kalman filter to obtain the important PDF, and solves the particle degeneracy and degradation of PF.

(4) SVM/RVM-based method

SVM has been firstly proposed by Boser et al. in 1992. It is good at dealing with the linear inseparability of sample data through penalty function and kernel function techniques. As one of the most representative methods in the field of machine learning, SVM has applications in pattern recognition, regression estimation and so on. Its optimization goal is to minimize structural risks and enhance its generalization ability [132]. Based on kernel function, nonlinear problem in low dimensional space can be transformed into linear problem in high dimensional feature space. Typically, diagnosis is based on some function defined on the input space, and learning is the process of inferring the parameters of that function. For example, Meng et al. [131] used a two-layer structure SVM with radial basis function (RBF) to accurately estimate the battery SOH. Among them, the introduced cost function plays a key role. However, SMV requires a large amount of computation in solving quadratic programming problems with a large enough amount of data, which will cause difficulties in practical application. Therefore, a least squares SVM (LS-SVM) which can transform the quadratic programming problem into a linear equation problem has been developed. It has to be mentioned that although this algorithm can make up for the deficiency of SVM, it loses the feature of sparsity. To solve these problems, Chen et al. [132] developed a fixed-size least squares SVM (LS-SVM) algorithm. The support vectors in the model were selected according to arbitrary entropy, and the parameters of the model were optimized by Bayesian framework. For calculation speed, LS-SVM has a great improvement over the traditional SVM.

Support vector regression (SVR) is a variant of SVM in dealing with regression problems. Based on SVM, Yang et al. [133] developed a method of least squares support vector regression (LSSVR) algorithm to estimate SOH. Firstly, a battery model combining OCV model and Thevenin model was established. Then the model parameters, OCV and SOC were identified by joint extended Kalman filter-recursive least squares (EKFRLS) algorithm. Finally, according to the obtained parameters, the SOH value was estimated by PSO-LSSVR method. Among them, the objective function J of the LSSVR algorithm can be expressed as Eq. (8):

$$\min J(w, \xi) = \frac{1}{2} w^T w + \frac{C}{2} \sum_{i=1}^l \xi_i^2 \quad (8)$$

where w represents the weight of a nonlinear transformation, ξ and C stand for the error variance and penalty factor respectively, l represents the number of samples.

RVM uses kernel functions less frequently than SVM and is capable of avoiding the use of free parameter sets [134]. RVM also has applications in SOH estimation. In literature [134], the mean entropy-based method was proposed to select the optimal embedding dimension for time series reconstruction, and the obtained optimal embedded dimension was regarded as the input variable of the RVM model which has been introduced the wavelet denoising method. The polynomial kernel function is used to map the original input to the high-dimensional feature space, which is defined as:

$$K(x, x_i) = (\langle x, x_i \rangle + 1)^d \quad (9)$$

where d can be deemed as scale parameters with controlling the resolution of a kernel.

3.2.3. Hybrid methods for SOH estimation

A hybrid method is a collection of two or more methods. Compared with those single model-based methods or data-driven methods, hybrid methods usually have higher accuracy. Therefore, the application of the hybrid method is introduced from four perspectives, including: the combination of direct measurements and model-based/data-driven methods, the mixture of data-driven and model-based methods, and the fusion of different data-driven methods.

SOH is estimated by combining the direct measurements with the model-based method. The combination of these two methods will obviously have a deeper exploration for the internal electrochemical mechanism of the battery. Xiong et al. [135] detected the degradation state of the battery using EIS measurements and abstracted the double polarization model from the EIS. Then the FFRLS algorithm was applied to identify the model parameters, and the correlation between parameter changes and degradation level was studied. Based on the exponential monotonic relationship between SEI resistance and residual capacity, the battery SOH was monitored. Furthermore, literature [136] used the forgetting factor recursive expanded least-squares (FFRELS) to identify fractional order model parameters online. By using ICA method to analyze the aging mechanism of batteries, it was found that there was a linear relationship between IC peak and residual capacity.

As for the combination of direct measurements and data-driven methods, the authors in [137] estimated SOH by combining EIS measurements with FL methods. The EIS measurement results of two LIBs with different aging processes were regarded as input parameters of the fuzzy logic system (FLS) to predict the SOC/SOH of the batteries.

The data-driven approach is combined with the model-based approach. For example, in literature [138], a multi-scale framework for SOH diagnosis was built based on the Brownian Motion (BM) degradation model and PF, which could realize online short-term SOH estimation. With a view to improve the precision of SOH monitoring, BM with non-linear drift and scaling parameters was used to infer the dynamic behavior of battery degradation. Song et al. [139] developed a joint SOH estimation method based on LS-SVM and unscented particle filter (UPF), which took the measurable terminal voltage and current as input. Considering the influence of noise, Sun et al. [140] used complementary ensemble empirical mode decomposition to process the error data between the estimated value obtained based on UPF and the real value. Meanwhile, the evolution trend of error was predicted based on the optimized multiple kernel RVM.

The hybrid methods between data-driven methods are the most common. This type of hybrid method does not require the internal mechanism and expert knowledge of the battery. Health diagnosis can be performed only through the historical data of the battery, which is more convenient than the previous several mixing methods. Chen et al. [141] developed a hybrid model based on autoregressive moving average (ARMA) model and Elman NN to estimate SOH. The empirical

mode decomposition (EMD) was used to cope with the capacity degradation data, so that three intrinsic mode functions (IMFs) and residual were gotten. Then, ARMA model and Elman NN were applied to estimate IMFs and residual respectively, and SOH prediction was made based on the estimation results. In literature [107], a method based on the fusion of multi-scale logistic regression (LR) and GPR was developed. Combined with time series prediction of different scales, the estimation of battery SOH was realized.

4. RUL prediction methods

Like SOH, RUL prediction is also a vital component of battery health management and an important reference index for evaluating the residual value of retired LIBs. In the stage of cascade utilization of retired batteries, only by grasping the information of remaining life can the batteries that are about to be aged and damaged be replaced regularly to ensure the safety of system and thus prevent the occurrence of safety accidents.

Because the RUL of battery is affected by many factors such as current health state, historical data, and failure mechanisms, it is hard to calculate the RUL. Predicting the battery RUL remains a challenge. As a result, more and more researches are being done on the RUL prediction of batteries. In view of the detailed introduction of model-based methods in SOH estimation methods, the application of model-based methods in RUL prediction is not summarized in this section. Interested readers can refer to references [142–154]. Combined with the research status at home and abroad, this chapter divides RUL prediction methods into two categories: data-driven method, hybrid methods, as shown in Fig. 12.

4.1. Data-driven method for RUL prediction

The data-driven prediction methods mainly predict battery RUL by summarizing the variation rules of the battery performance parameters based on the experimental data. These data contain all kinds of disturbances and influences of the system working environment on batteries. Hence, this kind of method can break through limitations of low dynamic accuracy and poor universality for model-based method to some extent. The main flow of prediction based on a data-driven approach is shown in Fig. 13. Firstly, different parameter data of the battery are collected by various sensors, and the corresponding algorithm is selected to preprocess the battery data. Then the extracted features are applied to train the established model, and the corresponding prediction results are finally obtained.

This chapter discusses data-driven prediction methods from four perspectives: time series-based method, artificial intelligence (AI)-based method, filtering-based method and stochastic process-based method. Meanwhile, applications of the above various RUL prediction methods are summarized in Table 10.

4.1.1. Time series-based method

Speaking of the method of predicting battery RUL based on time

series, the AR model and its variants are the most widely used methods. The ARMA, which consists of an AR model and a moving average (MA) model, is a commonly used model in time series analysis. Box et al. [155] proposed the autoregressive integrated moving average (ARIMA) model, which is a good prediction method for nonlinear stationary time series. The ARIMA model only considers a single variable, describes the battery degradation law through the capacity change, and then uses the law to predict RUL. Therefore, the application and performance of RUL prediction based on time series are discussed from the perspective of AR and its improved algorithm (ARIMA).

In view of the difficulty in determining the AR order and MA order in this method, AR model is usually combined with other optimization algorithms for RUL prediction. In literature [110], the AR model was employed to predict the capacity degradation of LIBs. The particle swarm optimization (PSO) algorithm was applied to search the optimal order of AR model, and the order was changed adaptively by metabolism updating information. In addition, long-term forecasting such as multi-step time series forecasting is a challenging problem, so some improved AR models have been developed. For example, in literature [156], the nonlinear acceleration degradation factor was extracted based on the analysis of the nonlinear capacity degradation. An improved AR model, named nonlinear degradation AR (ND-AR) model, was put forward by combining the nonlinear degradation factor with the standard AR time series prediction model. It was found that the improved model can get more accurate prediction results of RUL, and the model was also suitable for low power calculation. In order to improve the prediction accuracy of the model, a study introduced the iterative updating strategy in the ND-AR model and proposed an improved nonlinear degenerate AR (IND-AR) model [157]. The general flow of the forecast is shown in Fig. 14. Based on nonlinear degradation factor and iterative parameter updating method, the long-term prediction performance is improved.

The capacity decay process of the battery is usually nonlinear, while the AR model is linear. This discrepancy will cause the model under-fitting, especially for long-term prediction. However, ARIMA can compensate for these shortcomings of the AR model. Zhou et al. [158] developed a RUL prediction method combining EMD and ARIMA models. Firstly, EMD was used to decompose the degradation trend and capacity regeneration from the original time series, and the decomposed sequences were predicted based on the ARIMA model. The results of the predicted decomposition sequence were spliced to obtain the SOH prediction results, so as to indirectly predict RUL.

AR model and its improved models have low computational cost and complexity. However, the model is built solely on data, and the prediction results depend on the variation characteristics of the input data of the model, so the confidence interval of the prediction results is large.

4.1.2. Artificial intelligence-based method

The method based on AI can be applied to both SOH estimation and RUL prediction. SOH information is usually regarded as input for RUL prediction. This method mainly uses its nonlinear fitting ability to fit the variable degradation model on the data, and establishes the mapping relationship between the characteristic parameters and the battery RUL.

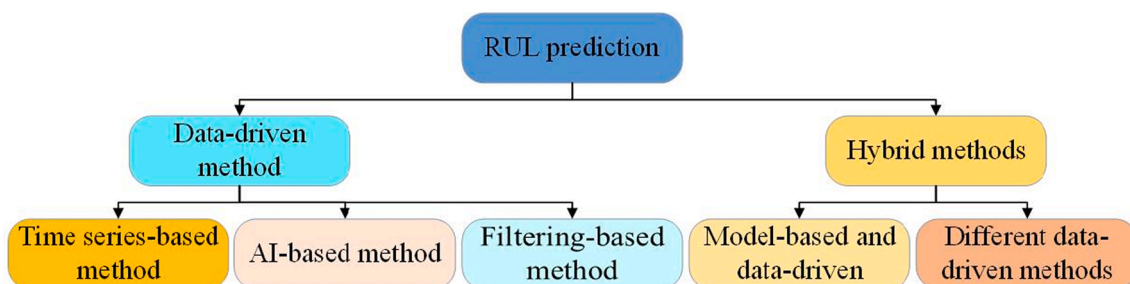


Fig. 12. Classification of RUL prediction methods.

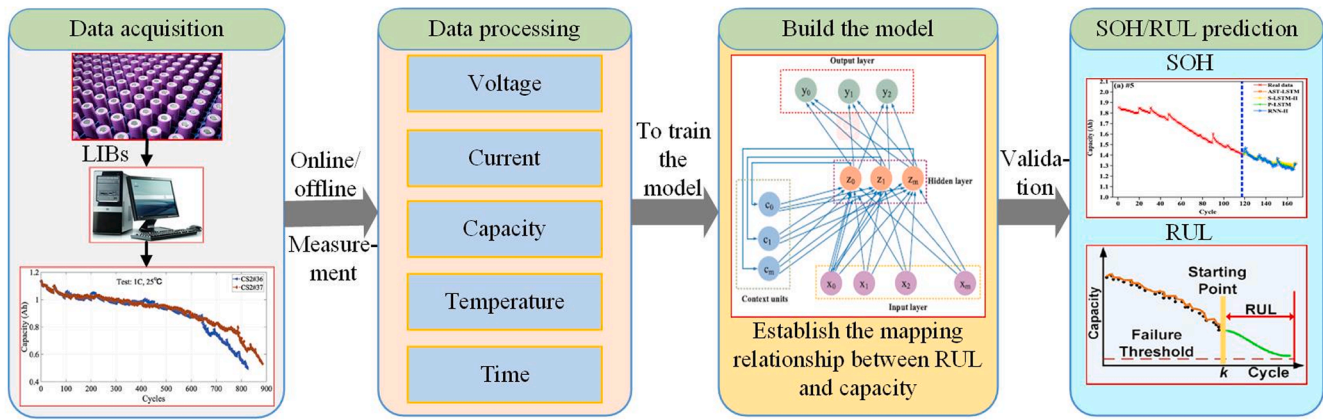


Fig. 13. Flowchart of the data-driven method for RUL prediction.

Table 10

The applications and evaluation results of data-driven for RUL prediction.

Methods	Refs.	Computational complexity	Evaluation index	Estimation precision
Time series-based method	AR	[110] Matrix operation, AR and PSO [157] Matrix operation, IND-AR and RPF [156] Matrix operation, ND-AR	Error Error MAE, RMSE, Error	0.123 < 8% < 0.7% (MAE), < 1.3% (RMSE), < 10% (Error)
AI-based method	ARIMA	[158] Matrix operation, EMD, ARIMA	RMSE	< 0.05
	NB	[159] Matrix operation, NB	RMSE, SD	0.17% (RMSE), 0.15% (SD)
	NF	[162] Matrix operation, NF, NN, RF, GMDH	RMSE, Error	N/A
	ANN	[165] Matrix operation, AdNN	MSE	0.9% (After 371st Cycle)
		[166] Matrix operation, CPNN	N/A	N/A
		[167] Matrix operation, FFNN	MAE, MSE	29.4218 (MAE), < 0.17% (MSE)
		[170] Matrix operation, ARNN, RLM	Accuracy	within 90%
		[171] Matrix operation, RNN, MLR	Error	1.43% (NCM523 of 25C), 0.21% (NCM622 of 30C)
		[172] Matrix operation, LSTM RNN	Error	N/A
	SVM	[174] Matrix operation, SVM	RMSE, Error	0.242% (RMSE), > 99% (Error)
		[175] Matrix operation, PCC PF-SVR	N/A	N/A
		[176] Matrix operation, FVS-SVR	RMSE	< 1.6%
	RVM	[178] Matrix operation, RVM, exponential functions	Absolute prediction error	< 10 cycles
		[179] Matrix operation, RVM, Box-Cox Transformation	Absolute error (AE), η AE	< 10 in most cases (AE, NASA)
Filtering-based method	KF	[180] Matrix operation, IP-RVM, R-RVM	AE, η AE	(6, 1) (IP-RVM), (2, 4) (R-RVM)
		[181] Matrix operation, DBN, RVM	MAE, RMSE, MRE	120unit numbers: (2.7-15.7); (0.29-1.05); (2.02-4.89)
		[182] Matrix operation, KF, RVM	Error, RUL_error, MAE, RMSE	4 (Error), 0.0312 (RUL_error), 0.0308 (MAE), 0.0014 (RMSE)
			RMS	< 3%
	EKF	[183] Matrix operation, EKF	RMS	< 0.065% (SOC)
		[184] Matrix operation, EKF	N/A	N/A
	DEKF	[185] Matrix operation, DEKF, ANN	RMS	< 1.08% (SOC), < 1.52% (SOH)
		[114] Matrix operation, DEKF, FFRLS	Error	A4 (18cycles): 0.9310 (RA), 0.1605 (MAPE), 0.01578 (RMSE), 0.9778 (R^2)
	UKF	[186] Matrix operation, UKF, RVR	RMSE, MAPE, R^2 , RA	B0018 (60cycles): 0.0233 (RMSE), 0.0121 (MAE), 0.8389 (R^2), 0.933 (RA)
		[187] Matrix operation, AUKF, GA-SVR	RMSE, MAE, R^2 , RA	
Stochastic process-based method	PF	[189] Matrix operation, PF, PDF	N/A	N/A
		[190] Matrix operation, exponential expressions, UPF	Error	< 5%
		[193] Matrix operation, EKF, PF	Error	< 5%
		[192] Matrix operation, exponential expressions, U-LOC-R-PF	AE, RPE, RMSE	70cycles: 1 (AE), 1% (RPE), 0.01983 (RMSE)
		[196] Matrix operation, SCPF	AE	< 6 cycles (Testing sample 6)
		[197] Matrix operation, PPF, KS	RMSE	< 4.5%
	GPR	[198] Matrix operation, dual GPR	MAE, RMSE	< 1% (SOH), < 23cycles (RUL)
		[199] Matrix operation, GPM, exponential expressions	RMSE	< 3.5%
	WP	[200] Matrix operation, MVDGPR	MAE, R^2	$R^2 > 0.9$ and MAE < 0.1
		[201] Matrix operation, WP, TND, MLE	MSE	PDF
		[202] Matrix operation, WP, MLE	N/A	PDF

According to the current research status at home and abroad, the AI-based methods are mainly divided into naive Bayes model, neuro-fuzzy model, ANN, SVM and RVM.

(1) Naive Bayes model

Naive Bayes (NB) algorithm is derived from classical mathematics and relies on conditional independence assumption. It is a simple classification approach based on Bayes' theorem. In all the given target values, the attribute values are independent of each other, and at the same time, the attribute variables account for a small proportion of the

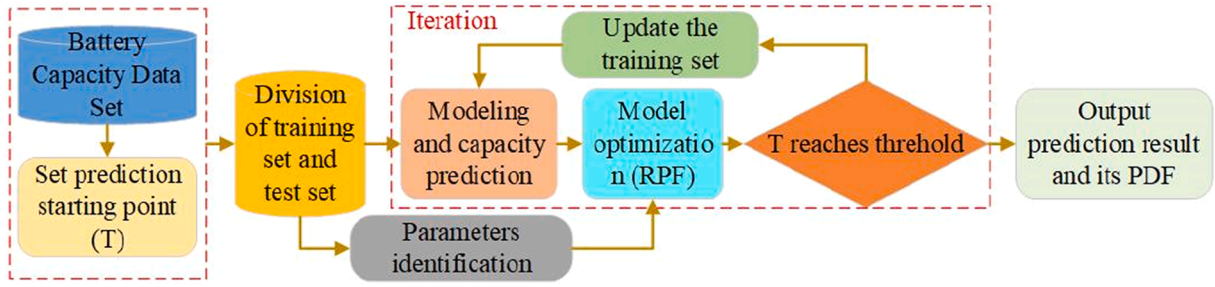


Fig. 14. The basic prediction process.

final decision result. Suppose there are a samples, and each sample includes n features, the feature output contains K categories and is defined as $\{C_1, C_2, \dots, C_K\}$. The conditional probability distribution and joint distribution of NB are shown in Eq. (10) and Eq. (11) respectively:

$$P(X = x|Y = C_K) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n|Y = C_K) \quad (10)$$

$$P(X, Y = C_K) = P(Y = C_K)P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n|Y = C_K) \quad (11)$$

In literature [159], the authors trained a NB model to predict RUL using the publicly available dataset. Considering the external conditions such as different ambient temperature and discharge current, the robustness of battery RUL under different constant conditions over time was quantitatively studied. The historical capacity data was regarded as the input variable, and the RUL value of different batteries was the output variable.

(2) Neuro-fuzzy model

Neuro-fuzzy (NF) model is developed on the basis of NN and FL. The fusion of these two makes up for the deficiencies of NN in fuzzy data processing and FL in learning. NF has strong approximate ability and transparent structure. As a common variant of NF, the adaptive neural fuzzy inference system (ANFIS) establishes the nonlinear mapping relation between the input and output of the system based on human experience and knowledge. Through systematic training and learning, the parameters related to MF of adjusting system variables are obtained, and a fuzzy reasoning system composed of several fuzzy rules is generated [160]. ANFIS can make inferences under uncertainty and predict the RUL of LIBs [161]. For example, Razavi et al. [162] investigated the robustness of four data-driven approaches (i.e., NNs, group method of data handling, NF and random forests) for battery RUL prediction. Three different predictors were constructed in NF model method, and it was found that ANFIS did not need expert knowledge to train input-output relationship, and the prediction accuracy of NF model was better than that of the other three methods.

(3) ANN

ANN is an AI network system formed by the regular connection of many neurons, and it is a non-linear method based on data [163]. Section 3.2.2 introduces the application of ANN for SOH estimation. It is impossible to ignore that ANN and its rich improved algorithm are also widely used in RUL prediction field. For instance, ANN model was employed to predict the capacity degradation of battery in Ref. [164]. According to the nonlinear autonomous learning function of ANN, massive data samples are used to train neurons, and the corresponding weights and thresholds are adjusted to determine the model.

Although ANN has a good prediction effect at the beginning of battery life, its prediction effect at the end of battery life is not satisfactory. Therefore, there are more and more researches on ANN optimization algorithm. Rezvani et al. [165] studied the application of adaptive neural network (AdNN) in battery RUL. Taking the battery capacity as the input variable, a capacity prediction model was established. It was

worth mentioning that the proposed method has good single-step prediction performance. In view of the shortcoming that ANN can only give a point estimate value, the confidence prediction neural network (CPNN) was introduced in literature [166] to predict RUL on the basis of the dynamic NN. The output of CPNN was the probability density distribution of RUL. Moreover, in literature [167], a RUL estimation method based on feed forward neural network (FFNN) (see e.g. Refs. [168,169]) was proposed. On the basis of analyzing the voltage curve under different cycle times, the relation between RUL and charge curve was simulated by FFNN. After the current cycle number of the battery is estimated by FFNN, the RUL value is calculated based on Eq. (12):

$$N_{RUL} = N_{EOL} - N_{ECL} \quad (12)$$

where N_{RUL} stands for the cycle number of RUL. N_{EOL} and N_{ECL} represent the cycle number when battery comes to its end of life and the equivalent circle life respectively. These studies show that the ANN optimization algorithm can achieve a good prediction effect in the whole life cycle of batteries, not just limited to the initial stage, and can effectively make up for the shortcomings of the traditional ANN algorithm.

RNN is a NN that connects nodes into a loop, and its internal state can display dynamic temporal behavior [16]. Since RNN can only conduct short-term capacity prediction and cannot achieve the purpose of long-term RUL prediction, Liu et al. [170] proposed an adaptive recurrent neural network (ARNN). ARNN weights were adaptively optimized by using the recursive Levenberg-Marquardt (RLM) algorithm. Several battery impedance data were used as model input to characteristic the impedance degradation trend, thus obtaining RUL value indirectly. In literature [171], the authors conducted accelerated aging experiments and EIS tests on NCM523 and NCM622 batteries with different design ratios. According to the data from the accelerated degradation experiment, the battery RUL was predicted by multiple linear regression and RNN. The LSTM variant of RNN has the ability to capture long-term dependencies between the battery degradation capacities. A data-driven predictor was built according to LSTM RNN [172]. Among them, LSTM RNN was trained by the resilient mean square back-propagation technique.

(4) SVM

SVM is able to overcome the disadvantages of ANN such as local extremum, complex network structure and large sample training. It can improve the generalization ability of the algorithm and has the ability to deal with finite small sample datasets [173]. It is worth noting that the computational cost of SVM is much lower than that of ANN. Therefore, SVM has been widely applied in the prediction field.

In order to ameliorate the prediction precision and save the calculation cost, Patil et al. [174] developed a real-time RUL prediction method based on two-level SVM. The classification technology was used to roughly estimate RUL, and then the regression technology was used to precisely predict RUL. Among them, on the basis of analyzing the cyclic data of the battery under different operating conditions, the minimum critical feature set was extracted from the voltage and temperature

distribution as the input of the classification and regression model. It is precisely because of the classification steps that the time-consuming regression model is only activated when the end of battery life is approaching, so this method greatly cut down the computation.

Compared with using SVM alone, there are more researches on the improved algorithm of SVM or the fusion of SVM with other algorithms. In literature [175], an SVR model combining PF with possibilistic clustering classification (PCC) was proposed to characterize the degradation process of batteries at different temperatures. PF and PCC were used for state estimation, and then SVR parameters were selected according to the system state to predict RUL under different operating states. In order to quantify the relationship between online measurable variables and capacity, a capacity prediction model combining feature vector selection (FVS) and SVR algorithm was established, which improved the efficiency of model construction and prediction [176].

(5) RVM

RVM is proposed on the basis of sparse Bayesian theory [177]. Compared with SVM, RVM not only uses fewer kernel functions and the number of associated vectors, but also it can provide probabilistic predictions [134]. RVM has received extensive attention in RUL prediction. In literature [178], the authors used RVM to indirectly estimate the battery RUL. A representative training vector was obtained based on RVM. Then a conditional three-parameter capacity degradation model was established to fit the predicted value of the selected vector.

It is worth noting that optimizing and improving RVM can often get better prediction results. In order to enhance the dynamic update capability, Liu et al. [179] developed an optimized RVM algorithm. To enhance the relevance between extracted HI and capacity and improve the prediction performance of RUL, the Box-Cox transformation was used to optimize HI. Then the optimized HI was regarded as the input of RVM to predict RUL. In another study [180], a new optimized incremental RVM model was proposed for long-term RUL prediction. The model obtained training data based on relevant vectors and online data, and considered incremental data samples. Therefore, in the face of long-term forecasting problems, this method can be used as an option to solve the relevant problems.

RVM is also often used in conjunction with other algorithms to predict RUL for higher accuracy. For example, according to the complementary principle, Zhao et al. [181] combined deep belief network (DBN) with RVM to predict battery RUL. DBN had good feature extraction capability, while RVM can just make up for the lack of uncertainty representation capability of DBN.

4.1.3. Filtering-based method

Filtering-based method is one of the common approaches for battery health monitoring. Filtering technology is mainly used to dynamically track and predict the health state of battery through the established state equation and the performance parameters. The commonly used filtering-based algorithms are divided into KF, PF and the improved algorithm of both. These filtering methods belong to Bayesian filtering family and offer a general framework for dynamic state estimation.

(1) KF algorithm

KF is advantageous in dealing with linear systems with Gaussian noise. The basic principle of KF: with the minimum mean square error as the best estimation criterion, the linear system state equation is used to estimate the system status according to the estimated value of previous time and the observed value of current time. Dynamic systems are usually described using the following state-space models:

$$X(k+1) = A \cdot X(k) + B \cdot U(k) + w(k) \quad (13)$$

$$Y(k) = H \cdot X(k) + v(k) \quad (14)$$

where k and $U(k)$ stand for discrete time and control vectors, respectively. $w(k)$ and $v(k)$ represent input noise and observation noise, respectively. A , B and H stand for the state matrix, control matrix and observation matrix respectively.

In the practical application of battery RUL prediction, the KF algorithm is usually used together with other algorithms for higher prediction precision. In literature [182], a fusion framework based on RVM and KF was proposed to predict battery lifetime. First, KF was used to optimize the estimator and update the training dataset. Then the updated dataset was applied to train RVM model twice. With the help of KF algorithm, better performance of RUL estimation was obtained.

1) **EKF**. Since the degradation process of LIBs is usually nonlinear, the KF algorithm cannot highlight its advantages in dealing with such problems. Therefore, an improved KF algorithm, EKF, has been proposed. The nonlinear system equation can be expressed as Eq. (15) and Eq. (16):

$$X(k+1) = f[k, X(k)] + w(k) \quad (15)$$

$$Z(k) = h[k, X(k)] + v(k) \quad (16)$$

where $f(\cdot)$ and $h(\cdot)$ stand for the nonlinear function of state equation and the observation equation, respectively.

In the literature [183], the battery SOC and capacity was estimated based on EKF. The developed framework combining Coulomb counting and adaptive filtering technology, which greatly saved the calculation cost and improved the accuracy. The output of the framework included SOH, so it can also be used to predict RUL. Bressel et al. [184] estimated the RUL of PEM fuel cells by degradation model and EKF. An empirical regression model was established based on the selected parameters. Then, EKF was used to estimate the time-varying parameters and their derivatives in aging process of batteries, and finally RUL was derived.

2) **DEKF**. DEKF algorithm combines the two EKF to work alternately. One EKF takes SOC as a state variable and considers capacity as a fixed value. The other EKF takes SOC as the input value and capacity as the state variable to obtain SOH change by predicting capacity decay. The principle of DEKF algorithm is shown in Fig. 15. DEKF is applicable to the estimation of nonlinear systems with simultaneous changes in parameters and states. Bai et al. [185] applied a method combining DEKF and ANN to monitor BMS. First, ANN was trained with offline data and updated adaptively according to online data. Then, based on DEKF algorithm, the trained ANN was applied to the state space equation to estimate SOC and SOH. Fang et al. [114] identified the parameters of ECM online using FFRLS algorithm, and then estimated SOC and SOH through DEKF algorithm according to the relation between battery resistance and SOH. The method developed based on DEKF can be further employed to accurately predict the battery RUL considering the long-term capacity decline.

3) **UKF**. UKF is an algorithm that combines UT transform with KF. The estimation accuracy of this method is higher than EKF, and it is a good nonlinear estimation method with a good development prospect.

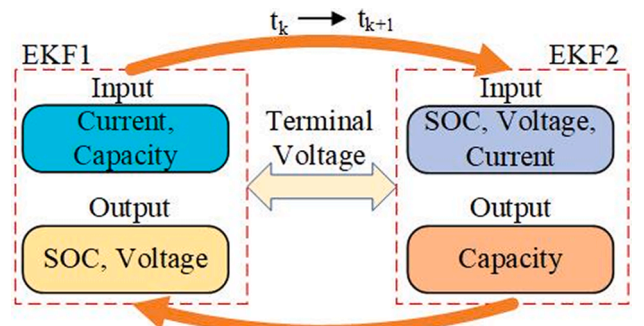


Fig. 15. DEKF algorithm schematic diagram.

Its nonlinear discrete system model can be expressed as Eq. (17) and Eq. (18):

$$X(k+1) = f(x(k), w(k)) \quad (17)$$

$$Z(k) = h(x(k), v(k)) \quad (18)$$

where $f(\cdot)$ and $h(\cdot)$ respectively stand for the nonlinear transformation function and nonlinear observation function of the system.

In the literature [186], the authors estimated RUL and short-term capacity through UKF and relevance vector regression (RVR), in which UKF was applied to estimate and adjust the system state to grasp the degradation process. Xue et al. [187] proposed a framework combining adaptive UKF (AUKF) and genetic algorithm optimized support vector regression (GA-SVR). AUKF was used to adaptively update the noise covariance, and then prediction of RUL was made by combining with SVR on this basis. It was worth noting that the filtering accuracy of AUKF was higher than that of standard UKF.

(2) PF algorithm

The PF algorithm based on Bayesian-Monte Carlo framework is an effective state tracking method for the nonlinear and non-Gaussian complex system such as battery. PF represents the posterior probability density function of a nonlinear system with a set of particle samples, so as to obtain a minimum variance estimate of the state process [188]. Hence, PF algorithm is widely applied in RUL prediction of battery (e.g. Refs. [189–197]). For instance, in the literature [189], based on the established lumped parameter battery model, the authors used the PF method combined with sequential importance resampling algorithm to achieve the battery RUL prediction.

Many improved PF algorithms have been developed due to the problems of important function selection and the diversity degradation in sampling particles for the standard PF algorithm. In literature [190], the authors proposed the UPF algorithm that used the results of UKF as the proposal distribution to predict RUL. Among them, the capacity estimation equation based on UPF and the probability density distribution of RUL prediction were defined as:

$$Q_k = \sum_{i=1}^N Q_k^i = \sum_{i=1}^N [a_k^i \cdot \exp(b_k^i \cdot k) + c_k^i \cdot \exp(d_k^i \cdot k)] \quad (19)$$

$$P(Q_{k+p} | Q_{0:k}) \approx \sum_{i=1}^N \omega_k^i \delta(Q_{k+p} - Q_{k+p}^i) \quad (20)$$

where a_k^i, b_k^i, c_k^i and d_k^i represent measurement functions, and ω_k^i is the normalized weights.

To further optimize the forecast performance, Zhang et al. [192] proposed a UPF algorithm according to linear optimizing combination resampling (LOCR), where UKF was employed to generate an important function for PF to overcome particle degradation to some extent. Then, the LOCR algorithm was applied to overcome deficiency of particle diversity. In literature [193], the authors used EKF as a sampling density function to optimize PF. The extended Kalman particle filter (EKPF) algorithm was developed to predict RUL based on the second order exponential capacity degradation model. Comparative experiments showed that the prediction precision of EKPF was higher than that of standard PF.

Spherical cubature particle filter (SCPF) is an improved PF that uses the importance function provided by spherical cubature integration-based KF (SCKF) to update the weight of random particles. The filter can provide more weights with the recent capacity data to update the importance function, thus achieving a better prediction effect than standard PF. For instance, Wang et al. [196] introduced SCPF to solve the established state-space model (SSM) for estimating battery capacity degradation. The determined SSM was extrapolated to the fault

threshold to predict RUL. In order to solve the problem that particle weights cannot be updated and particle degradation, Zhang et al. [197] developed an algorithm that combines F-distribution PF (FPF) and kernel smoothing (KS). Based on the advantages of FPF and KS, the RUL prediction performance of the proposed algorithm was good.

4.1.4. Stochastic process-based method

Stochastic process belongs to the category of statistical theory and can be divided into non-stationary and stationary stochastic processes. The degradation process of LIBs shows strong randomness and uncertainty, so it can be used to describe the degradation state of LIBs. In general, stochastic process includes gaussian process regression (GPR) and wiener process (WP).

(1) Gaussian process regression

GPR was proposed by Williams et al in 1996. The principle of this method is to train the prediction model with historical samples, and then use the test samples as the model input to get the prediction results with probability significance. Compared with ANN, SVM and other methods, GPR is suitable for nonlinear systems with higher dimension, and the hyper-parameters of its kernel function are more easily obtained.

Compared with most algorithms that can only make short-term predictions, GPR is able to make long-term predictions for batteries. Li et al. [198] estimated the SOH and RUL through the dual GPR model. The correlation coefficient approach was employed to analyze the correlation between the feature extracted from the local area of IC curve and the capacity degradation. Then, the feature with high correlation was selected as the input variable of the GPR-based battery degradation model for short-term SOH estimation. Finally, the SOH estimation results and previous outputs were regarded as the input of the autoregressive GPR-based model to obtain long-term RUL prediction. In view of self-charging and capacity regeneration, the capacity degradation trajectory of LIBs has multimodality. To overcome this problem, Li et al. [199] developed a RUL prediction framework based on gaussian process mixture (GPM), which used different GPR models to fit different track segments of capacity curve. Compared with the single GPR model, the proposed method can fit different capacity degradation trajectory and thus has better prediction performance.

The deep learning algorithm is of great help to improve the prediction accuracy. A recent study proposed a deep learning-based method by using the matrix-variate deep GPR (MVDGPR) [200]. Due to the advantages of its deep architecture, this method can directly take part of the charging and discharging time series data as input variables for capacity estimation and further prediction of RUL. It is precisely because of its deeper structure that it can achieve higher prediction accuracy than traditional GPR, but it also increases computational complexity.

(2) Wiener process

Wiener process (WP) is a typical stochastic process in which the increments show a steady characteristic. The change increment of WP at any time period follows a normal distribution, and its variance increases linearly with time.

Because random processes are irregular, WP can be studied based on probability theory and statistics. Therefore, the probability distribution of RUL can be attained by using WP model. Tang et al. [201] developed a RUL prediction method based on WP with measurement error. Based on truncated normal distribution (TND) and maximum likelihood estimation (MLE), the accurate probability distribution of RUL was obtained. In another study [202], the authors extracted two HIs indirectly based on the charge profile information, and established a two-dimensional WP model to characterize the battery degradation process. After using the MLE algorithm to estimate the parameters, the PDF of RUL was obtained. In addition, there are also more and more applications of WP in combination with other methods. In literature [203], the authors used

WP with three state variables of random drift, diffusion coefficient and measurement error to characterize the capacity degradation process of LIBs. Subsequently, the RUL prediction of secondary battery was attained based on WP and PF algorithm.

4.2. Hybrid methods for RUL prediction

In recent years, hybrid methods have been research hotspots for both SOH estimation and RUL prediction. The hybrid model formed by combining or integrating multiple algorithms can make up for deficiencies of a single model and make full use of the advantages of different algorithms so as to attain better performance. The hybrid approaches are divided into two types: the fusion of model-based and data-driven methods and the fusion of different data-driven methods. The application and evaluation results of these two types of hybrid methods are summarized in Table 11.

4.2.1. The fusion of model-based and data-driven methods

A popular idea in model-based and data-driven hybrid methods is to combine data-driven algorithms with state-space models. More accurate prediction results can be obtained through combining the characteristics of the aging model with the real-time state information.

Most prediction algorithms based on PF rely on physical model or empirical model, so the research of PF fusion with other algorithms is extensive. For example, in literature [204], the authors proposed a PF framework that integrates the model and data-driven approach. The Levenberg-Marquardt method was used to optimize the empirical model, which was established to characterize the battery degradation trend. Then the PDF of RUL was obtained by using PF on this basis. Combining the merits of both data-driven and model-based approaches, Liu et al. [205] proposed a fusion prediction approach based on AR and PF. In order to enhance robustness of prediction and the uncertainty representation ability of PF, ND-AR model was used as the observation equation of regularized PF.

In the process of battery degradation, filtering algorithms are usually used to update model parameters to maintain predictive performance. However, initialization and sample impoverishment are two major

problems in filtering algorithms. The initial values are usually set based on experience or randomly [17]. If an improper initial value is set, the convergence speed of the algorithm will decrease, or even divergence. He et al. [25] initialized PF parameters through Dempster-Shafer theory to improve convergence speed. In order to prevent particle degradation, traditional PF uses resampling technology to eliminate small-weight particles while copying large-weight particles, but this will lead to sample impoverishment [17]. To avoid particle degradation, Zhang et al. [206] used Markov Chain Monte Carlo (MCMC) method to maintain the diversity of particles, and the proposed improved UPF (IUPF) could also inhibit particle degradation.

There are two challenges with current battery RUL prediction methods: a large amount of training data is usually required to provide model learning and the performance of the algorithm cannot be effectively verified. In view of these two challenges, Zhang et al. [207] proposed a fusion framework of RVM-PF. Based on this framework, only 30% of online degradation data was needed to adaptively update model parameters. Meanwhile, the Monte Carlo method was introduced as a benchmark to verify the prediction performance of PF. Furthermore, there is a universal long-range dependence between degradation data of the actual battery system. A non-Markovian fractional Brownian motion (FBM) model was adopted to effectively describe long-range dependence and fruit-fly optimization algorithm (FOA) for accurate RUL prediction [208].

4.2.2. The fusion of different data-driven based methods

Combining two or more data-driven methods by weighting or other means can improve the robustness of the predicted results. This kind of hybrid method is widely used.

With the rapid development of machine learning, it is becoming more and more common to integrate different ANNs into a new prediction model to predict the battery RUL. For instance, a recent study [209] predicted the battery RUL by combining LSTM-RNN and Elman NN. Based on these two networks, the sub-models of the obtained low frequency capacity sequences are established respectively. LSTM-RNN can capture the evolution trend of long-term signals based on its cyclic link, while Elman NN can describe the dynamics of short-term signal fluctuations. Therefore, the fusion framework improves the prediction accuracy under both long and short windows. In another example [210], based on the false nearest neighbors method, the authors made full use of the feature extraction ability of CNN and the time series prediction ability of LSTM to achieve RUL prediction. Among them, the structure of CNN-LSTM was composed of two convolution layers, one LSTM layer and one fully connected layer.

The combination of AI-based method and mathematical-based method can also achieve good prediction performance. Wang et al. [211] developed a hybrid model according to SVR and differential evolution (DE). Considering the performance parameters (i.e. capacity, voltage and current) in the battery discharge process, the kernel parameters of SVR were obtained by using DE. Zhao et al. [212] proposed a hybrid prediction model based on RVM and grey model (GM). More robust prediction results were obtained by iterative update strategy.

In general, prediction of RUL based on PF algorithm depends on the selected empirical model. However, if the empirical model cannot accurately describe the battery degradation trend, the predicted results will be inaccurate. To overcome this problem, Wu et al. [213] replaced the empirical model with NN that is more flexible and has more computational power to simulate various degradation trends of the battery, and used bat-based PF algorithm to update the weights and deviations of NN. Experimental results showed that the proposed method improved the prediction precision of RUL.

5. Analysis and comparison

So far, the research status of SOH estimation and RUL prediction has been clarified as far as possible. For the field of SOH estimation, this

Table 11
The applications and evaluation results of hybrid methods for RUL prediction.

Method	Refs.	Computational complexity	Evaluation index	Estimation precision
Fusion of model-based and data-driven methods	[204]	Matrix operation, PF, empirical model	RMSE	< 3.5%
	[205]	Matrix operation, ND-AR, RPF	MAE, RMSE, Error	0.0335 (MAE), 0.0246 (RMSE), 9 (Error)
	[206]	Matrix operation, IUPF, MCMC	RMSE_Mean, RMSE_Variance	18 cycles: < 0.3% (Mean), < 0.045% (Variance)
	[207]	Matrix operation, RVM, PF	N/A	N/A
	[208]	Matrix operation, FOA, FBM	MSE, RMSE	RW12: <1.3 (MSE), <1.2 (RMSE)
Fusion of different data-driven methods	[210]	Matrix operation, CNN, LSTM	EM, AM, SM	< 6 cycles (EM)
	[209]	Matrix operation, LSTM RNN, Elman NN	Error	< 3.3%
	[211]	Matrix operation, SVR, DE	MAPE, RMSE	< 0.15% (MAPE), < 0.1% (RMSE)
	[212]	Matrix operation, RVM, GM	AE, RMSE	< 15 cycles (AE), < 3.5% (RMSE)

paper divides the methods into direct measurement and indirect analysis. The direct measurement method is primarily to directly measure HIs of the battery based on the internal resistance test, impedance test or other test methods. Although these types of methods do not need to use a lot of data and the calculation is small, its precision is low and it is not fit for online estimation. Indirect analysis methods include model-based method and data-driven method. The model-based approach is summarized in more detail. For the RUL prediction field, since the process of using model-based approach to predict RUL is similar to that in SOH estimation, thereby this part only introduces the data-driven method and hybrid method in detail. In a word, there are two dominant methods in the fields of SOH estimation and RUL prediction, which are model-based method and data-driven method respectively. In recent years, applications of hybrid method are increasing. The hybrid approach combines advantages of model-based with data-driven approaches, and researchers are increasingly inclined to use hybrid approaches. Fig. 16 shows the SOH estimation and RUL prediction methods summarized in this paper.

Model-based approaches require less data and are less sensitive to external uncertainties, so they are more stable. But the model-based approach relies on expert knowledge to develop physical models, and the modeling process is complex. The data-driven method is more flexible and has high prediction accuracy. Nevertheless, it is susceptible to data uncertainty and incomplete, so its generalization ability and adaptability are poor. Hybrid method can greatly enhance the prediction performance and become a popular development trend in the future. However, the method still has some limitations such as high computational cost and uncertainty fusion. Table 12 lists the advantages and disadvantages of model-based, data-driven and hybrid approaches, respectively.

Table 12

The model-based method, data-driven method and hybrid method for evaluation.

Method type	Advantages	Disadvantages
Model-based method	1. Low computational effort 2. Prediction stability 3. Possible implementation in a BMS	1. Not suited for situ estimation 2. Need to know the exact electrochemical reaction 3. Accurate physical or chemical models are needed
Data-driven method	1. Suitable for offline monitoring 2. High prediction accuracy 3. No physical or chemical model is required	1. Prediction performance is unstable 2. High computational effort 3. The prediction accuracy depends on the mathematical model
Hybrid method	1. High prediction accuracy 2. High practical value	1. High computational effort 2. Uncertainty fusion

5.1. Evaluation of model-based method

The model-based evaluation methods require the use of professional knowledge and experience to analyze the battery and establish the aging model from the perspective of the essential mechanism of electrochemical reaction. Table 13 summarizes the advantages and disadvantages of various model-based approaches.

The construction of ECM is highly realizable, but the implicit relations between the parameters of the battery may be neglected in the approximation process, which may play a decisive role in the system characteristics. At the same time, it is difficult to fully consider the complex external conditions. Therefore, ECM is weak in describing the dynamic and static characteristics of the battery.

Detailed physical and chemical analysis of the aging process can be

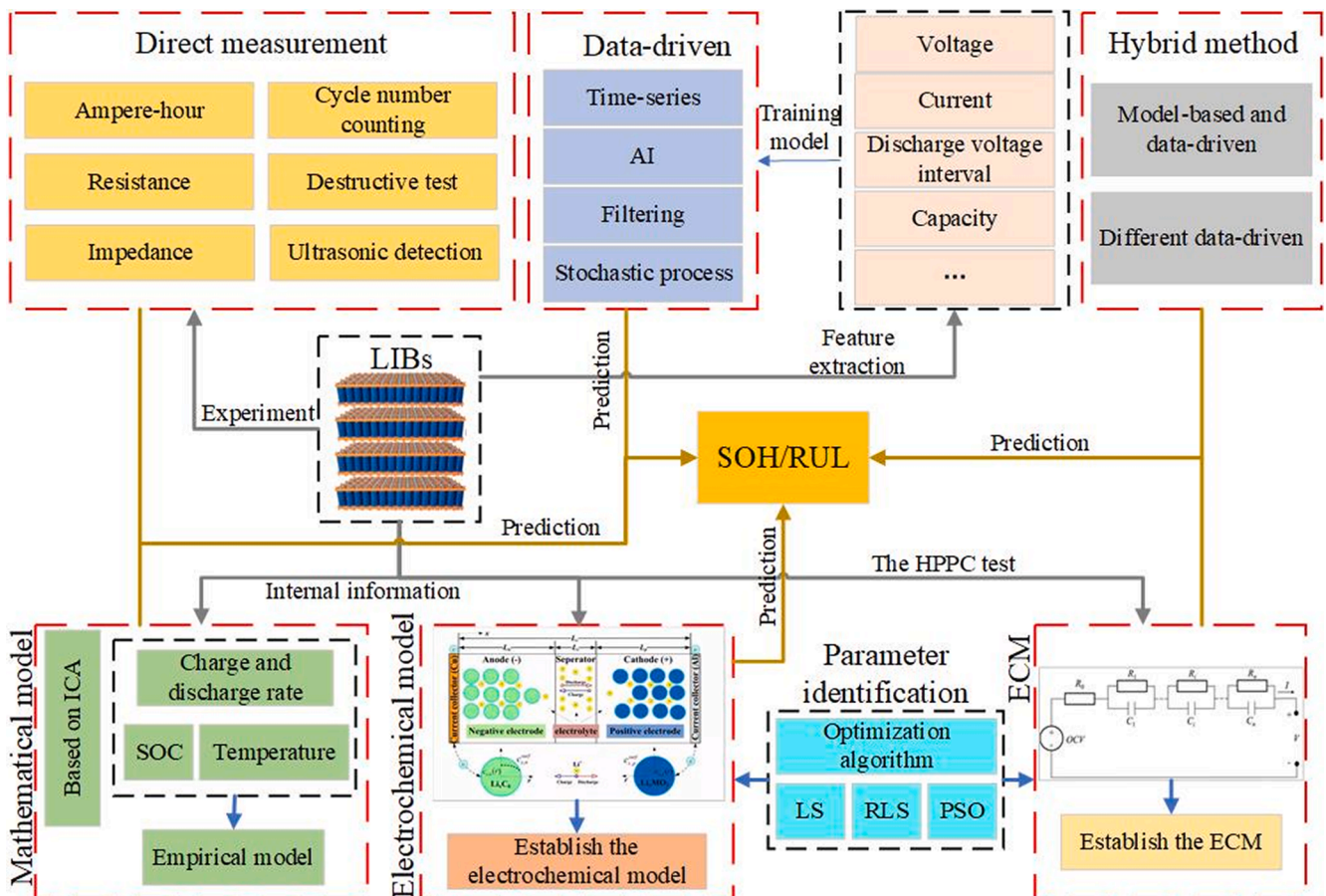


Fig. 16. The methods of SOH estimation and RUL prediction.

Table 13
Evaluation of model-based methods.

Method	Advantages	Disadvantages
ECM	1. In line with the actual use 2. The chemical composition and reactions inside the battery are not considered	1. High cost 2. The aging process is accelerated by EIS test
EM	1. High prediction accuracy 2. A detailed explanation of battery degradation can be given	1. The model is complex 2. Degradation factor testing is difficult
Mathematical model	1. Easy to establish 2. Strong adaptability	1. Low prediction accuracy 2. Empirical knowledge is required

obtained by the EM-based method. However, EM depends on specific battery materials and the charging-discharging conditions. The model parameters are usually attained through the physical characteristics of the electrode. Therefore, the dynamic accuracy of EM is poor. In addition, EM has higher complexity and more model parameters.

The mathematical model is easier to establish and has great adaptability to non-gaussian and nonlinear degradation process. However, its dynamic precision and adaptability are still limited due to the fact that the mathematical model is unable to take the environmental disturbance, load dynamic characteristics and other influencing factors into consideration.

5.2. Evaluation of data-driven method

Data-driven evaluation methods include time series, artificial intelligence, filtering and stochastic process-based methods. All kinds of methods have different applicable scope and can obtain good prediction effect under certain conditions. At the same time, different complexity and operating conditions affect the application of various methods. The advantages and disadvantages of each method are listed in Table 14.

AR model is a classical time series method, which has good real-time prediction performance and is suitable for small sample cases with low calculation cost. However, the prediction accuracy of this method is low and it is not suitable for nonlinear signals.

The five methods are summarized in the AI-based method, which are NB, NF, ANN, SVM and RVM respectively. They can deal with more complex situations, and have relatively high precision. However, these methods rely on data characteristics and are computationally complex.

The methods based on filtering are divided into KF and PF. Among them, KF has good real-time prediction performance and prediction accuracy, but it has the limitation of high computing cost. The ability to express uncertainty is a major advantage of PF. However, the initialization process is complex and the prediction process requires a high-precision degradation model.

GPR and WP are two of the most common stochastic process-based methods. Both of these methods have the ability to express uncertainty. GPR is a powerful tool for coping with problems of small sample and high latitude regression. WP is useful for dealing with non-monotone modeling problems. However, their calculations are relatively complex.

6. Challenges and future trends

At present, many researchers are working on developing new battery materials and processes. In particular, the development and research of high safety and high energy density batteries are more concerned. However, research in the related fields of status monitoring and health management is still in its infancy. To inspire more technological innovation and transformative breakthroughs, as shown in Fig. 17, we try to delineate the roadmap for battery health management and discuss future challenges.

Table 14
Evaluation of data-driven methods.

Method	Advantages	Disadvantages
Time series-based method	AR 1. Good real-time performance 2. Appropriate for small samples 3. Low computational effort	1. Low accuracy 2. Point estimates 3. Not applicable to nonlinear signals
AI-based method	NB 1. Insensitive to missing data 2. Concise NF 1. ANN is used to optimize the inference structures 2. Can solve inaccuracy and fuzzy characteristics ANN 1. Strong independent learning ability 2. High algorithm accuracy SVM 1. The global optimal solution can be obtained 2. High prediction accuracy RVM 1. Few parameters and high sparsity 2. Arbitrary kernel function 3. Ability to express uncertainty	1. Based on the assumption of attribute independence 1. High computational effort 2. Adequate training data is required 1. Lack of ability to express uncertainty 2. Not appropriate for small samples 1. Kernel and regularization parameters are difficult to calculate 2. Lack of ability to express uncertainty 1. Poor stability 2. Not appropriate for long-term prediction
Filtering-based method	KF 1. Suitable for real-time processing 2. High prediction accuracy PF 1. Probabilistic prediction 2. Ability to express uncertainty	1. Method is complex as it requires large amount of calculations 1. The initialization process is complex 2. The establishment of degradation model should be accurate
Stochastic process-based method	GPR 1. Suitable for regression problem of small sample and high dimension 2. Strong adaptability WP 1. Ability to express uncertainty 2. Suitable for non-monotonic process modeling	1. Large amount of calculation 2. Hyperparameter adjustment is complex 1. Does not apply to nonlinear and time-inhomogeneous data 2. Based on Markov property

- (1) How to provide the high-quality data for training models and solve the small sample problem?

With the widespread use of LIBs, the amount of the collected data is rapidly increased than ever before. However, the quality of the entire data is not often satisfactory because a part of them may be poor quality. Therefore, it is necessary to develop valid methods to clean up erroneous data. Clustering algorithms and reasoning models can help eliminate abnormal data and further improve the quality of collected data. In addition, the crowdsourcing database technology can efficiently manage data with low value density that helps establish standard databases to provide high-quality data for predictive models. On the other hand, in the case of few circulating data and single working condition, small sample problem is often generated when using machine learning theory to conduct LIBs model, which has a significant impact on the generalization ability and performance of the model. Transfer learning is a new way to solve small sample problem. Based on the idea of knowledge transfer, any battery related to the target battery can be used for auxiliary modeling to make up for the missing battery information under the condition of small sample and break the distribution limitation between data through knowledge transfer.

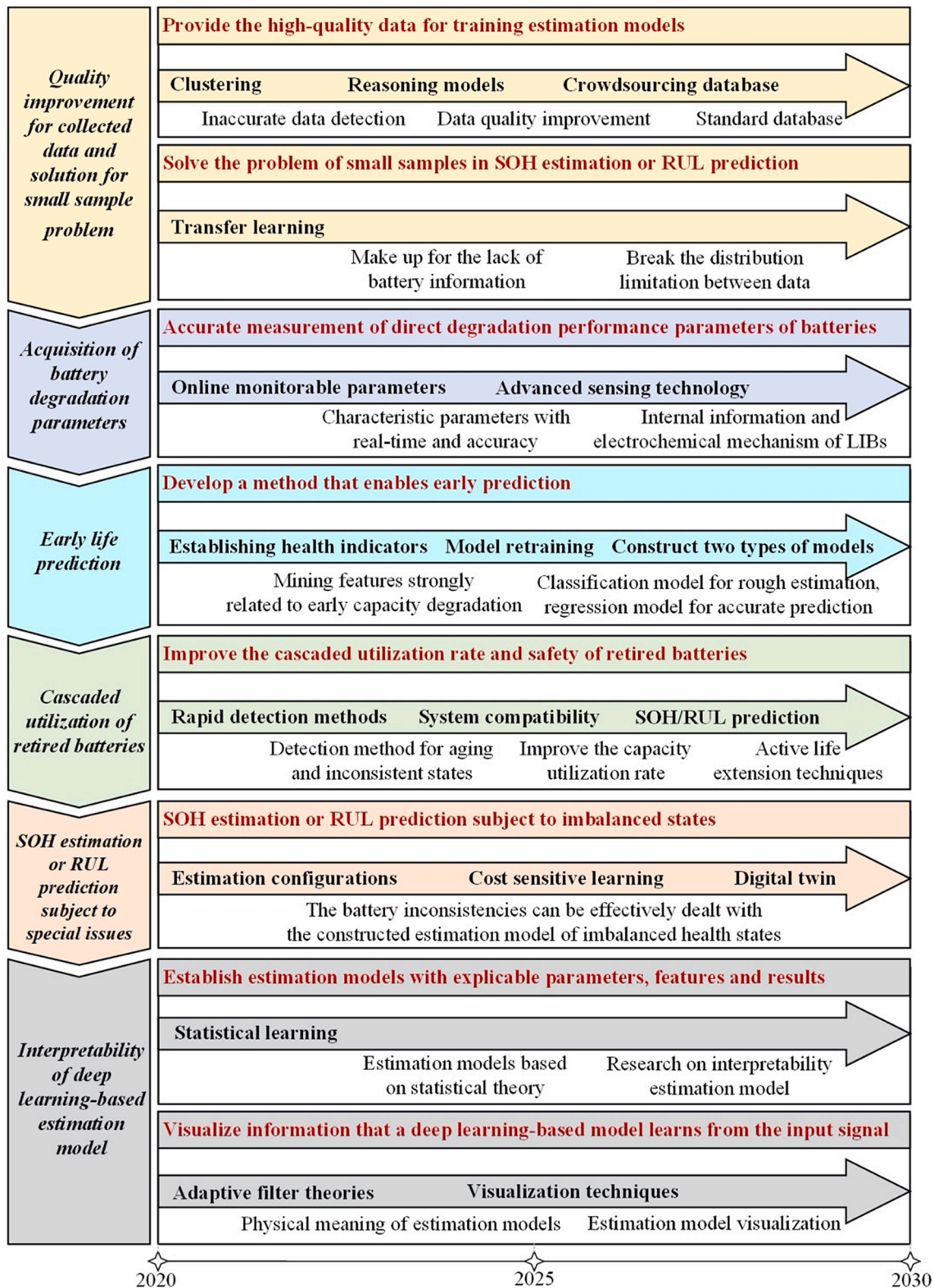


Fig. 17. Roadmap of applications of intelligent battery health management.

(2) How to accurately measure the degradation parameters of battery?

In the actual operating system, the direct performance degradation parameters of batteries can hardly be accurately obtained online. Therefore, in order to achieve the online estimation and prediction of

battery performance, it is necessary to focus on the online monitorable performance parameters. The characteristic parameters, such as time and rate of change, which are easily measured, can be extracted from the online monitorable performance parameters. These parameters are real-time and accurate, which can fundamentally improve the prediction accuracy and online evaluation ability. Furthermore, there is no doubt

that accurate measurement of the physical and chemical reactions in the battery with appropriate sensors is the most direct and accurate way to obtain the state parameters. It is urgent to develop advanced sensing technology. High-precision contact-type displacement sensors have been used to measure the force generated by cell expansion [214,215]. As a typical advanced sensor, the optical fiber sensor can monitor additional battery parameters, which has the advantages of strong anti-electromagnetic interference ability and sensitive parameter selection. In the future, the improvement of sensing technology can be started from the efficient acquisition of key information inside the battery and in-depth understanding of internal electrochemical mechanism.

(3) How to achieve the early RUL prediction of batteries?

For most RUL forecasting methods, being able to predict RUL accurately at an early stage is a huge challenge. Therefore, developing a method that can realize early prediction is crucial for accelerating the development of battery technology and preventing battery failure. Inspired by the work [216], three possible solutions to the early prediction of RUL are proposed. The first method is to mine the characteristics in the original data that have strong correlation with the early capacity degradation and establish the relevant health indicators. However, the applicability of the selected indirect health factors in different working conditions needs to be verified in many aspects. The second approach is to train the prediction model offline with available data and then retrain the prediction model with a small portion of online data [17]. The third method can be inspired from the research in literature [216]. According to the available data, the classification model can be employed to roughly predict RUL in the early period, and then the accurate prediction of RUL can be obtained by the regression model. All in all, the algorithms that provide early prediction with limited available data still needs further study.

(4) How to improve the cascaded utilization rate of retired batteries?

Promoting the cascade utilization of retired batteries and giving full play to the surplus value of batteries are undoubtedly of great practical significance to the battery industry. However, most of the cascade utilization projects of LIBs are still in the demonstration or experimental stage. Therefore, large-scale promotion of cascade utilization of retired LIBs still faces many challenges. Based on some challenges of the cascaded utilization, three recommendations are put forward. First, in terms of detection and screening, more effective detection and evaluation methods should be designed based on single battery technology according to aging and inconsistent states of cascade battery modules. Rapid detection methods, such as EIS technique, should be further developed. Second, the inconsistencies under electric and thermal stress should be fully considered in the recombination of cascaded batteries, and the compatibility of battery system to different kinds of cascaded batteries should be improved based on the technical aspects of battery layout and power distribution, so as to improve the capacity utilization rate of cascaded batteries. Thirdly, considering that the life of cascaded battery is in the middle and late stage, the aging mechanism of battery should be taken into full consideration to carry out the research on life-cycle SOH estimation and RUL prediction algorithm. In addition, active life extension techniques can be added.

(5) How to perform SOH estimation or RUL prediction at module and package-level?

In battery-powered equipment, it is usually necessary to use a battery pack composed of a large number of cells in series or in parallel to meet specific power requirements. Health diagnosis of module-level and pack-level needs more attention. However, due to differences in materials and manufacturing parameters between cells, as well as temperature gradients and uneven aging between cells [104], it is difficult to establish a

pack aging estimation model. In view of these problems, three suggestions are proposed. First, different estimation configurations should be proposed according to the estimation requirements of different system levels. Interclass correlation coefficient-based method can effectively solve the imbalance among cells. For example, Li et al. [217] successfully eliminated the problems of voltage sensors noise and internal inconsistency based on interclass correlation coefficient method. Second, the cost sensitive learning can help to conduct estimation models for health state imbalance. In addition, ensemble learning such as Adaboost and XGboost, combined with resampling strategies, is expected to improve the performance of deep learning-based estimation models. Third, digital twin, as an emerging technology, is expected to provide some solutions to these problems. According to establish the digital twin of cell to generate the aging trajectory of cell under different working conditions, the cell inconsistency, changes in environment and working conditions can be effectively dealt with.

(6) How to enhance the interpretability of deep learning-based estimation model?

Although the estimation model based on deep learning has achieved relatively advanced results, the black box problem of deep hierarchical networks is still an inexplicable problem. It is difficult to master the process of learning prediction knowledge from monitoring data, so estimation models are constructed through multiple experiments rather than rigorous theoretical basis. Based on this, two research directions are suggested to be paid attention to. First, the physical meanings of the estimation model based on deep learning are hard to explain. The statistical learning theories, such as the probabilistic graphical model, have strict theoretical principles, which is of great help to construct an estimation model whose model parameters, features and estimation results are easy to explain and understand. As a result, it is worthwhile to study the statistical learning in intelligent estimation for battery state with the context of the big data revolution. Secondly, the adaptive filtering theory is helpful to analyze the physical meaning of estimation model based on deep learning. Furthermore, the visualization techniques such as feature inversion should be further developed to visually describe the information that the estimation model learns from the input signals.

7. Conclusion

In this paper, the development of SOH estimation and RUL prediction techniques for LIBs is analyzed in detail. Firstly, the commonly used public battery datasets are introduced. The SOH estimation methods based on direct measurement and model are mainly reviewed, and the latest progress of RUL prediction based on data-driven and hybrid methods is summarized in detail. Then, various methods are evaluated. Their advantages and disadvantages are analyzed and thoroughly discussed. Finally, some recommendations are made for the existing challenges.

The degradation of LIBs is a complex process. Although there are many ways to estimate SOH and predict RUL, this field is still in its infancy. The model-based method does not require much data, but its complexity is high and it needs expensive experimental equipment. The data-driven method is more flexible and has high prediction accuracy, but it has the disadvantage of large computation. The hybrid method can combine the advantages of model-based and data-driven approaches to provide more precise prediction. It is worth noting that these methods all show good prediction performance in ideal experimental environment, but most of them are not suitable for practical engineering application. The following three points can be taken as the starting point of future research:

- (1) Due to the inconsistencies in the battery pack, digital twin technology should be further applied in the field of life prediction for pack.

- (2) Since battery aging data are difficult to measure, transfer learning should be further used to deal with the prediction problem with a small data sample.
- (3) In order to facilitate the reuse of retired batteries, the RUL algorithm for second-life should be further developed.

With the wide application of LIBs, more and more attention will be paid to the development of SOH estimation and RUL prediction methods for LIBs which can be applied to practical engineering with good performance. Through this article, the authors hope to contribute to the development of SOH estimation and RUL prediction algorithms for LIBs, and hope that anyone interested in this area will benefit from this work. It is worth mentioning that since each method has its own shortcomings and the possibility of improvement, there is not a perfect method for SOH estimation or RUL prediction of LIBs. The choice of approach needs to consider many factors, such as user requirements, data availability, and availability of LIBs degradation mechanisms. All in all, through the above review, considering the complexity of the LIBs system and working conditions, the advantages of machine learning-based methods will become more and more prominent. Hybrid methods, which complement each other in the shortcomings of model-based and data-driven methods, are also promising in the field of SOH estimation and RUL prediction for LIBs. Although this paper attempts to make a comprehensive summary of SOH estimation and RUL prediction for LIBs, some perspectives have still not been fully discussed. In the future, other aspects such as the aging mechanism of LIBs and methods of extending battery life need to be thoroughly explored.

CRediT authorship contribution statement

Ming-Feng Ge: Methodology. **Yiben Liu:** Writing - original draft, Data curation, Software, Validation. **Xingxing Jiang:** Writing - review & editing, Investigation, Visualization. **Jie Liu:** Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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